

Price Risk, Irreversible Investment, and the Welfare Benchmark for the Section 45Z Clean Fuel Production Credit

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August 2026

We decompose the welfare case for the Section 45Z clean fuel production credit into its two statute-relevant components, the carbon externality and price risk under irreversible investment, and ask whether a flat per-gallon credit is the right instrument for the risk component. A two-stage stochastic mixed-integer program sites a forest-residue sustainable aviation fuel (SAF) supply chain across 1,374 southeastern U.S. counties, and a closed-form benchmark calibrated to the same price distribution supplies the analytics. That distribution is a kernel density fitted to a constructed drop-in fuel price: slate-weighted wholesale gasoline–diesel plus the Renewable Fuel Standard credit value the fuel earns, weekly over 2013–2021. Price risk justifies at most \$1.04/gal at any admissible risk weight, and \$0.52/gal at the midpoint calibration, so risk alone cannot cover the statutory \$1.30–\$1.45/gal. The balance implies a social cost of carbon of \$77–\$91/tCO₂ under the as-enacted rate and flat-credit delivery. Committing capacity before prices are known costs \$0.30 B/yr in expected regret, about a fifth of the chain’s \$1.43 B/yr surplus, concentrated in the lowest-price scenarios. A one-sided price floor restores risk-neutral benchmark output at 69% of the flat credit’s expected outlay, and a two-sided contract-for-difference restores it at zero. The flat credit is the most expensive instrument for the risk component.

Keywords: sustainable aviation fuel; stochastic programming; welfare analysis; CVaR; social cost of carbon; contracts-for-difference.

JEL codes: Q42, Q23, D81, H23, C61, R12.

1. Introduction

Sustainable aviation fuel (SAF) requires large, irreversible investments in feedstock logistics and refining capacity, and future fuel prices are uncertain. SAF deployment is therefore a problem of investment under uncertainty. Consider the southeastern United States, the region most often discussed as the platform for forest-residue-to-SAF deployment. There, a producer would face a drop-in fuel price, inclusive of the Renewable Fuel Standard credits the fuel earns (Section 4.3). Its estimated support is \$3.81–\$9.12/gal. The planner must commit depot and biorefinery capacity before any of that price is observed.

U.S. policy now offers substantial per-gallon support for SAF through Section 45Z of the Inflation Reduction Act (IRA), with statutory credit values in the range of \$1.30–\$1.45/gal.¹ The statute scales the credit to carbon intensity, but the informal case for SAF support invariably adds a second argument: “de-risking” irreversible investment against volatile fuel prices. This paper prices both arguments in one framework. We decompose the capacity-restoring credit into a Pigouvian carbon component and a price-risk component, show the decomposition is exactly additive, and then ask a question the level-setting debate skips: whether a flat per-gallon credit is the right *instrument* for the risk component at all.

Existing work addresses parts of this problem but not the welfare question. Spatially detailed siting studies (Ekşioğlu et al. 2009; Lin et al. 2014; Marvin et al. 2012; Masum et al. 2023) represent feedstock heterogeneity, logistics, and processing costs richly but treat uncertainty deterministically or through scenario analysis. The stochastic-programming literature (Birge and Louveaux 2011; Schultz 2003; Shapiro, Dentcheva, and Ruszczyński 2014; Santos et al. 2019; Gebreslassie, Yao, and You 2012; Osmani and Zhang 2017; Quddus et al. 2018) supplies the framework for here-and-now decisions, committed before the price is known, but it is rarely implemented at regional spatial scale or in a welfare setting. Approaches based on the conditional value-at-risk (CVaR) (Rockafellar and Uryasev 2000, 2002; Artzner et al. 1999; Conejo, Carrión, and Morales 2010; Boomsma, Meade, and Fleten 2012; Insley 2017) price downside tail risk coherently, but none pairs CVaR with a large-scale siting problem under empirical price uncertainty. The one comparison of renewable support instruments under price uncertainty (Boomsma, Meade, and Fleten 2012) does not connect to the welfare accounting of a specific statutory credit. We join these strands.

We formulate SAF supply-chain investment as a social planner’s welfare-maximization problem with a make-or-buy choice between domestic SAF and wholesale fossil jet fuel at \$3.50/gal, well below the roughly \$6.00/gal mean of the policy-inclusive SAF revenue price. We solve it as a two-stage stochastic mixed-integer linear program (SP-MILP) over 1,374 counties in the southeastern United States. Because a deep SAF spot market does not yet exist, the uncertain revenue price is a *constructed drop-in biofuel price*: a slate-weighted wholesale gasoline–diesel price plus the market value of the Renewable Fuel Standard credits the fuel generates, weekly from August 2013 through December 2021 and summarized by a kernel-density estimate (KDE). That level embeds the credit value, so the credit we price is incremental to the existing compliance stack (Section 4.3). A closed-form benchmark maps the same price distribution into a welfare-equivalent per-gallon subsidy. It delivers two results: an exact additive decomposition of the capacity-restoring credit into Pigouvian and risk components

¹Section 45Z pays base rate $\times (50 - \text{CI})/50$ per gallon, with CI the pathway carbon intensity in kgCO₂e/mmBTU and a wage-and-apprenticeship-compliant SAF base rate of \$1.75/gal as enacted in the IRA (Congressional Research Service 2025b; Lohawala, Toman, and Joiner 2024). The \$1.30–\$1.45 range corresponds to a forestry-residue Fischer–Tropsch (FT) pathway CI of roughly 8.6–12.9 kgCO₂e/mmBTU under GREET-type accounting (Argonne’s Greenhouse gases, Regulated Emissions, and Energy use in Technologies life-cycle model).

(Proposition 1), and a strict expected-fiscal-outlay ranking across three instruments for the risk component, a flat credit, a one-sided price floor, and a two-sided contract-for-difference (CfD) (Proposition 2).

We report three findings. First, the welfare cost of committing capacity before prices are known is concentrated in the low-price tail. For the committed central plan, expected regret is \$299 M/yr, about a fifth of the chain’s \$1.43 B/yr surplus, carried entirely by the nine lowest-price scenarios. The 5%-tail is worse: \$921 M/yr, roughly two-thirds of that surplus. A planner with risk weight above roughly 0.10 builds a chain one-fifth smaller.

Second, calibrating the closed form to the empirical price distribution yields a risk-premium subsidy of $\sigma^* = \$0.52/\text{gal}$, and the gap to the statutory \$1.30–\$1.45/gal does not close at any admissible risk weight: the risk component tops out at \$1.04/gal under maximal tail aversion. Add the Pigouvian component, a life-cycle differential of 10.2 kgCO₂e/gal at the make-or-buy margin. The statutory credit is then consistent with an implied social cost of carbon (SCC) of \$77–\$91/tCO₂ under the baseline reading, between the two federal SCC vintages: \$51/tCO₂ from the Interagency Working Group (IWG) ([Interagency Working Group on Social Cost of Greenhouse Gases 2021](#)) and \$190/tCO₂ from the U.S. Environmental Protection Agency (EPA) ([U.S. Environmental Protection Agency 2023](#)). The implied SCC is a schedule rather than a point, and one row is news: the post-2025 rate actually in force prices carbon at \$22–\$30/tCO₂, *below* the IWG vintage.

Third, the flat credit is the most expensive instrument for the risk component. A one-sided price floor at \$5.64/gal restores the same capacity at 69% of the flat credit’s expected fiscal outlay, saving \$0.48 B/yr at the 3.0 B gal/yr anchor volume and up to \$0.14 B/yr in deadweight loss when each dollar of public funds costs society \$1.30. A two-sided CfD at an actuarially fair strike restores it at zero expected outlay, moving the price risk from the committing region onto the federal balance sheet ([Arrow and Lind 1970](#)). Pay less, and pay it differently.

The analysis is partial equilibrium; Section 6.1 signs the omitted channels, which move the risk component in both directions. The producer-side counterpart of the instrument-form result is developed in a companion paper ([Li and Yu 2026](#)). Section 2 develops the welfare framework and the closed-form benchmark, Section 3 the empirical model, and Section 4 the data and calibration; Section 5 reports the results, Section 6 discusses scope and limitations, and Section 7 concludes.

2. Conceptual Framework

A social planner chooses where to build a forest-residue-to-SAF supply chain in the southeastern United States, and at what scale, before SAF prices are realized. We organize the problem in four steps: the economic environment and timing (Section 2.1); the welfare objective with a make-or-buy outside option (Section 2.2); risk aversion and the value of information (Section 2.3); and a closed-form benchmark that maps the same primitives into a risk-premium-equivalent per-unit subsidy (Section 2.4). The closed form supports two further results used in Section 5: an exact additive decomposition of the capacity-restoring credit into a Pigouvian component and a risk component (Proposition 1), and an expected-fiscal-outlay ranking across the instrument forms that can deliver the risk correction (Proposition 2).

2.1. Economic environment and timing

We first fix notation for the geography, the sources of price and quality uncertainty, and the two-stage timing of the planner’s decisions. Let $\mathcal{H}$ index residue-source counties, $\mathcal{D}$ candidate *depot* counties (with biomass

pre-processing capability), and $\mathcal{R}$ candidate *biorefinery* counties. A scenario $\omega \in \mathcal{S}$ realizes a SAF wholesale price $p_{\text{SAF}}(\omega)$ and a feedstock-quality shock $\xi(\omega)$, with KDE-implied probability p^ω (proxied by historical wholesale jet-fuel spot prices, [U.S. Energy Information Administration 2024](#)). Following [Li et al. \(2026\)](#), we discretize feedstock ash content into percentile-based tiers $\mathcal{A}_d, \mathcal{A}_r$ at depots and refineries, with the SAF yield at refinery ash configuration $a' \in \mathcal{A}_r$ written $\phi(a'; \omega)$.

Decisions arrive in two stages. In Stage 1, the planner commits binary facility variables $y \in \{0, 1\}^{|\mathcal{D}|+|\mathcal{R}|}$. The scenario ω then realizes, and in Stage 2 the planner chooses continuous residue and pre-processed flows $\mathbf{x}(\omega), \mathbf{z}(\omega) \geq 0$ together with imported jet-fuel slack $s(\omega) \geq 0$. The Stage 1 choice is therefore made here-and-now, before ω is realized; Stage 2 captures flow-level recourse, the adjustment made after ω is observed.

2.2. Welfare objective and make-or-buy structure

Let $Q \in \mathbb{R}_{\geq 0}$ denote the year-zero representative output (gallons of SAF). For an inverse demand $P^d(Q; \omega)$, consumer surplus is

$$(1) \quad \text{CS}(Q; \omega) = \int_0^Q P^d(q; \omega) dq - P^d(Q; \omega) \cdot Q,$$

and producer surplus is revenue minus all chain costs,

$$(2) \quad \text{PS}(\mathbf{x}, \mathbf{z}, y; \omega) = p_{\text{SAF}}(\omega) \cdot Q(\mathbf{z}; \omega) - C_{\text{tot}}(\mathbf{x}, \mathbf{z}, y),$$

where $Q(\mathbf{z}; \omega) = \sum_{(d,a,r,a')} \beta \cdot \phi(a'; \omega) \cdot z_{d,a,r,a'}$ is total SAF output and C_{tot} stacks stumpage, harvest, transport, and facility opex and capex. In the endogenous-demand variant the realized price is the demand height at the equilibrium quantity and $\text{CS} + \text{PS}$ is total surplus; in the fixed-demand model P^d is horizontal, $p_{\text{SAF}}(\omega)$ is exogenous, and slack imports meet the demand floor (Section 3.5).

When domestic output falls short of demand, the gap is met by imported wholesale jet fuel at price p^{jet} , a perfect substitute within the blend limits of the ASTM International jet-fuel standard. Letting $s(\omega) \geq 0$ denote slack imports, the social opportunity cost of slack is $p^{\text{jet}} \cdot s(\omega)$; we calibrate $p^{\text{jet}} = \$3.50/\text{gal}$ as a long-run benchmark from 2014–2024 U.S. Energy Information Administration (EIA) Gulf-Coast jet-fuel data, with sensitivity in Section 6.1. The outside option prices unmet SAF demand, which is what makes the build-versus-no-build margin well defined. The planner’s problem is

$$(3) \quad \max_{\substack{y \in \{0,1\}^{|\mathcal{D}|+|\mathcal{R}|} \\ \mathbf{x}, \mathbf{z}, s \geq 0}} \mathbb{E}_\omega \left[\text{CS}(\mathbf{z}; \omega) + \text{PS}(\mathbf{x}, \mathbf{z}, y; \omega) - p^{\text{jet}} \cdot s(\omega) \right],$$

subject to the mass-balance, capacity, quality, and demand-satisfaction constraints of Section 3.

remark (Carbon externality). *On a consistent life-cycle basis, the fossil-jet baseline of CORSIA (the Carbon Offsetting and Reduction Scheme for International Aviation), 89 gCO₂e/MJ (International Civil Aviation Organization 2019), converts at 131.5 MJ/gal to 11.7 kgCO₂e/gal, while residue-derived Fischer–Tropsch SAF emits 1.1–1.5 kgCO₂e/gal (8.3 gCO₂e/MJ at the CORSIA default core value for forestry-residue FT; up to ~11 in GREET-type accountings, Argonne National Laboratory 2024). The differential is $\Delta e \approx 10.2$ – 10.6 kgCO₂e/gal; we use the conservative $\Delta e = 10.2$ throughout. The Pigouvian term $\text{ENV}(\mathbf{x}, \mathbf{z}, y; \omega) = -p_C \cdot E(\mathbf{x}, \mathbf{z}, y; \omega)$, with*

$E(\cdot)$ the chain's life-cycle emissions and p_C the carbon price, and the Section 45Z credit are both deferred: we set $p_C = 0$ in the stochastic program so that the price-uncertainty channel is isolated, and price the carbon channel analytically in Section 5.5.

remark (Welfare framing versus producer net present value). *At the price-taker limit $P^d(Q; \omega) = p_{SAF}(\omega)$ and $p_C = 0$, problem (3) reduces to the producer-side net-present-value (NPV) problem of [Ekşioğlu et al. \(2009\)](#); [Lin et al. \(2014\)](#); [Marvin et al. \(2012\)](#), except for the make-or-buy term $-p^{jet} \cdot s(\omega)$, which makes the build-versus-no-build decision non-trivial and delivers the public-finance object σ^* of Section 2.4.*

2.3. Risk aversion and information value

If the planner evaluating the regional welfare aggregate is risk-averse over the tail of its distribution at confidence level α (agent identities are fixed in Section 2.4), the expectation in (3) is replaced by the mean-CVaR functional

$$(4) \quad (1 - \lambda) \cdot \mathbb{E}[W] + \lambda \cdot \text{CVaR}_\alpha[W], \quad \lambda \in [0, 1],$$

where $W = \text{CS} + \text{PS} + \text{ENV}$ is total welfare. The functional is coherent ([Artzner et al. 1999](#)) and admits the [Rockafellar and Uryasev \(2000\)](#) linear-programming (LP) linearization; the empirical solves of Section 3 are risk-neutral, and the mean-CVaR program is evaluated on candidate plans rather than solved (Section 5.4). The functional prices tail risk in *welfare levels*; the per-scenario regret $L_s = W_s^{\text{WS}} - W_s^{\text{HN}}$ of Section 5.2 is a different object that concentrates in the same low-price states.

Three benchmark values organize the information question. Let $W^*(\omega) = \max W(\mathbf{x}, \mathbf{z}, y; \omega)$ be the per-scenario wait-and-see optimum, W^{RP} the two-stage recourse-problem optimum with y shared across ω , and W^{EEV} the expected value of the expected-value (EEV) solution, the objective when y is fixed at its mean-scenario optimum and $(\mathbf{x}, \mathbf{z})$ adapt. The expected value of perfect information (EVPI) and the value of the stochastic solution (VSS) are

$$(5) \quad \text{EVPI} = \mathbb{E}[W^*(\omega)] - W^{\text{RP}}, \quad \text{VSS} = W^{\text{RP}} - W^{\text{EEV}}.$$

EVPI is the planner's willingness to pay for perfect SAF-price forecasting; VSS is the cost of replacing the scenario distribution with its mean. Both are evaluated on candidate plans in Section 5.4 and Section D.

2.4. Closed-form benchmark: risk-premium-equivalent subsidy, decomposition, and instrument form

This closed-form benchmark is what generates the paper's headline per-gallon subsidy estimate, reported in the abstract, and shows the economics behind it. The large-scale program in (3) delivers a numerical optimum but leaves that mechanism implicit. A two-state reduction admits a closed-form solution that isolates how the welfare-equivalent per-unit subsidy depends on the upside probability, the demand-state spread, and the risk-aversion weight. The same machinery delivers the additive decomposition (Proposition 1) and the instrument ranking (Proposition 2).

Setup. Reduce (3) to a single output decision $Q \in \mathbb{R}_+$ in a two-state economy with demand state $\omega \in \{H, L\}$ and probabilities (π_H, π_L) summing to one. State-conditional inverse demand is linear, $P^d(Q; \omega) = w(\omega) - bQ$

with $b > 0$, and the intercept $w(\omega) \in \{w_H, w_L\}$ captures high/low realizations of the SAF willingness-to-pay distribution. Marginal supply cost is a constant c , and the representative-investor capacity cost is $C(Q) = \frac{k}{2} Q^2$ with $k > 0$. Writing $K \equiv b + k$ for the combined curvature, the area-under-demand-minus-cost integral gives

$$(6) \quad W(Q; \omega) = (w(\omega) - c) Q - \frac{K}{2} Q^2,$$

and the mean-CVaR functional in (4) applies directly.

Assumptions. This subsection and the proofs in Section A use the following maintained assumptions throughout.

assumption 1 (Two-state demand). $\omega \in \{H, L\}$ with $\pi_H, \pi_L > 0$ and $\pi_H + \pi_L = 1$.

assumption 2 (Linear inverse demand). $P^d(Q; \omega) = w(\omega) - b Q$ with $b > 0$.

assumption 3 (Convex capacity cost). $C(Q) = \frac{k}{2} Q^2$ with $k > 0$.

assumption 4 (State ordering). $w_H > w_L$.

assumption 5 (Positive net premium). $w_L > c$: even in the bad state the planner finds it worthwhile to produce.

assumption 6 (CVaR confidence). $\alpha \in (0, 1)$. The closed-form $\sigma^*(\alpha)$ is piecewise on $(0, \pi_H)$ and $[\pi_H, 1)$.

assumption 7 (Bounded risk-aversion weight). $\lambda \in [0, 1]$.

assumption 8 (Partial equilibrium). $w(\omega)$ is treated as an exogenous random variable; no general-equilibrium substitution between SAF and fossil jet fuel is modeled.

assumption 9 (External financing). Policy transfers are financed from outside the welfare aggregate over which the mean-CVaR functional (4) is evaluated: a credit enters the marginal return of the committing agent but its funding does not enter the CVaR argument.

Assumption 9 fixes whose risk the functional prices and who pays for the correction. Were the credit financed *inside* the aggregate, it would net out state by state and leave the program unchanged. Two readings satisfy it. In the regional reading, the functional is evaluated over the Southeast regional aggregate and the credit is a federal transfer into a small open region; this is the reading that matches the empirical model. In the decentralized reading, the tail-averse agent is the investor committing irreversible capacity; this is the reading developed in the companion paper (Li and Yu 2026). The algebra is identical under either, and we say “the committing agent” where the distinction matters. The normative benchmark is the risk-neutral objective $\mathbb{E}[W]$ of the external financier, whose taxpayer pooling supports risk neutrality, while λ describes the risk-bearing capacity of the committing agent, not the social planner’s preference. In plain terms, σ^* compensates for whose budget absorbs the loss, not for an unpriced externality: it is best read as a *jurisdictional risk-bearing wedge* rather than a Pigouvian market failure.

Microfoundation. Market incompleteness, not planner risk aversion, is why λ enters the welfare objective at all. Assumption 9 is the reduced form of that friction: the state-contingent claims that would let the committing agent sell its low-price exposure do not exist at the relevant tenor. Three frictions drive this. First, liquid hedging in the heating-oil/jet complex runs two to three years against a ten-year operating commitment on a twenty-year asset. Second, the premium of the produced fuel over jet-A has no settlement index. Third, a single-asset project company faces margin-call liquidity constraints that make a decade-long short position infeasible at scale (Froot, Scharfstein, and Stein 1993; Holmström and Tirole 1998). Under complete markets λ would be irrelevant and $\sigma^* = 0$; $\lambda > 0$ measures the shadow price of the missing claims, the committing agent’s allocation is constrained inefficient relative to the financier’s feasible set, and a correction raises welfare in that constrained sense, which is the sense in which we call the credit “capacity-restoring.” The first-best is to complete the market, Proposition 2(iii)’s two-sided contract, rather than to transfer resources; the flat credit’s σ^* is the level-equivalent benchmark against which the statute is read, not the recommended instrument.

Main result.

LEMMA 1 (Strict concavity and interior optimum). *Under Assumptions 1–7, the mean–CVaR welfare functional (4) evaluated at the per-state welfare (6) is strictly concave in Q on $\mathbb{R}_+$ with second derivative $-K < 0$, and the maximizer Q_{risk}^* is interior. Proof in Section A.*

THEOREM 1 (Risk-premium-equivalent subsidy). *Under Assumptions 1–9, the per-unit social subsidy σ^* that restores the risk-neutral output level Q_{RN}^* is*

$$(7) \quad \sigma^*(\alpha) = \lambda \cdot (w_H - w_L) \cdot \min\left\{\frac{\alpha \pi_L}{1 - \alpha}, \pi_H\right\}.$$

In the collapsed regime $\alpha \geq \pi_H$, which includes the baseline $\alpha = 0.95$ at $\pi_L \approx 0.5$, the expression simplifies to

$$(8) \quad \sigma^* = \lambda \pi_H (w_H - w_L).$$

Proof in Section A.

COROLLARY 2 (Expected value of perfect information). *Under Assumptions 1–8, the EVPI for the risk-neutral planner equals*

$$(9) \quad \text{EVPI} = \frac{\text{Var}(A)}{2K} = \frac{\pi_H \pi_L (w_H - w_L)^2}{2K}.$$

Proof in Section A.

COROLLARY 3 (Continuity at the regime boundary). *$\sigma^*(\alpha)$ is continuous on $(0, 1)$, with a kink in the derivative but no jump in the value, at $\alpha = \pi_H$. Proof in Section A.*

Economic interpretation. In the collapsed regime (the case where CVaR collapses to the low-price state’s welfare level rather than averaging over part of it), $\sigma^* = \lambda \cdot \pi_H \cdot (w_H - w_L)$ is the product of the committing agent’s risk-aversion weight, the upside probability, and the demand-state spread, each of which maps to an observable: λ is the weight a regulator selects when evaluating the exposed jurisdiction’s tail, π_H is calibrated

from the SAF-price KDE, and $w_H - w_L$ is the conditional-mean spread of its median split. The result extends the prices-versus-quantities benchmark of [Weitzman \(1974\)](#) to a mean-CVaR planner and is the discrete-scenario counterpart of the closed-form feed-in tariff of [Boomsma, Meade, and Fleten \(2012\)](#); at $\lambda = 0$ it recovers Weitzman's no-subsidy benchmark $\sigma^* = 0$.

Extension 1: additive Pigouvian decomposition. Section 45Z is a carbon-intensity-scaled production credit, so a welfare accounting of its level must handle the carbon externality and the risk premium jointly. In the mean-CVaR framework the two are exactly additive.

PROPOSITION 1 (Additive credit decomposition). *Suppose each unit of output carries an external social benefit $v \geq 0$ (the monetized value of avoided life-cycle emissions) not reflected in the market intercept $w(\omega)$. Under Assumptions 1–9: (i) the first-best output is $Q_{\text{FB}}^* = (\mathbb{E}[A] + v)/K$; (ii) the per-unit credit that moves the mean-CVaR planner to Q_{FB}^* is*

$$(10) \quad \sigma_{\text{tot}}^* = v + \sigma^*(\alpha),$$

with $\sigma^(\alpha)$ exactly as in Theorem 1; and (iii) the decomposition has no interaction term: the risk component is invariant to v , and the Pigouvian component is invariant to $(\lambda, \alpha, \pi_H, w_H - w_L)$. Proof in Section A.*

Additivity follows from the linearity of the effective intercept in the state-independent shift v : a constant per-unit externality raises the expectation and the tail conditional mean by the same amount, so it moves the target output without changing the risk wedge. The Pigouvian component can therefore be priced off the life-cycle differential and a social cost of carbon and added to the KDE-calibrated σ^* without double counting (Section 5.5).

Extension 2: instrument form. A flat credit shifts the mean of the payoff distribution while leaving its variance intact, whereas the distortion it corrects is a variance phenomenon. Instruments that operate on the variance deliver the same output correction at strictly lower expected fiscal outlay.

PROPOSITION 2 (Instrument form and expected fiscal outlay). *Let $m_\alpha \equiv \min\{\alpha\pi_L/(1 - \alpha), \pi_H\}$, so that $\sigma^*(\alpha) = \lambda(w_H - w_L)m_\alpha$ by Theorem 1. Under Assumptions 1–9, each of the following instruments restores the risk-neutral output $Q_{\text{RN}}^* = \mathbb{E}[A]/K$, with expected per-unit fiscal outlay T as stated:*

(i) Flat credit at rate $\sigma^*(\alpha)$, paid in every state: $T_{\text{flat}} = \sigma^*(\alpha)$.

(ii) One-sided price floor (CfD-with-floor) at strike

$$(11) \quad \kappa^* = \frac{\pi_L w_L + \lambda m_\alpha w_H}{\pi_L + \lambda m_\alpha} \in (w_L, w_H),$$

paying $\kappa^ - w(\omega)$ per unit only when $w(\omega) < \kappa^*$:*

$$(12) \quad T_{\text{floor}} = \sigma^*(\alpha) \cdot \frac{\pi_L}{\pi_L + \lambda m_\alpha} < T_{\text{flat}}.$$

(iii) Two-sided contract-for-difference at strike $\bar{w} \equiv \mathbb{E}[w(\omega)]$, exchanging the stochastic margin for the deterministic one ($\bar{w} - w(\omega)$ per unit, negative in the high state): $T_{\text{CFD}} = 0$, for every $\lambda \in [0, 1]$ and $\alpha \in (0, 1)$.

Table 1: Comparative statics for σ^* and EVPI in the collapsed regime $\alpha \geq \pi_H$.

Partial	Closed form	Sign	Interpretation
$\partial\sigma^*/\partial\lambda$	$\pi_H (w_H - w_L)$	+	Higher committing-agent risk aversion lifts the subsidy.
$\partial\sigma^*/\partial\pi_H$	$\lambda (w_H - w_L)$	+	Higher upside probability lifts the subsidy.
$\partial\sigma^*/\partial(w_H - w_L)$	$\lambda \pi_H$	+	Wider demand spread lifts the subsidy.
$\partial\sigma^*/\partial\alpha$ (split)	$\lambda \pi_L (w_H - w_L)/(1 - \alpha)^2$	+	Below the kink, tighter tail focus lifts the subsidy.
$\partial\sigma^*/\partial\alpha$ (coll.)	0	0	Above the kink, the subsidy is α -independent.
$\partial\text{EVPI}/\partial \text{Var}(A)$	$1/(2K)$	+	More demand uncertainty raises information value.
$\partial\text{EVPI}/\partial K$	$-\text{Var}(A)/(2K^2)$	-	Steeper curvature dampens information value.

For $\lambda \in (0, 1]$ the ranking is strict: $0 = T_{\text{CfD}} < T_{\text{floor}} < T_{\text{flat}} = \sigma^*(\alpha)$. Proof in Section A.

COROLLARY 4 (Welfare wedge under costly public funds). *At marginal cost of public funds $\mu = 1$ the three instruments of Proposition 2 are welfare-equivalent (transfers net out of the planner’s objective), and the ranking concerns expected public outlay only. At $\mu > 1$ the outlay differential becomes a strict welfare wedge: instrument i dominates instrument j by $(\mu - 1)(T_j - T_i) Q_{\text{RN}}^*$ per period. Proof in Section A.*

The economics is Arrow–Lind risk reallocation (Arrow and Lind 1970). The two-sided CfD transfers the price risk from the tail-averse committing agent to the external financier, so the wedge disappears without any expected transfer. The one-sided floor concentrates payments in the low states that drive the CVaR term, which is why it undercuts the flat credit by $\pi_L/(\pi_L + \lambda m_\alpha)$. The flat credit, paying identically in every state, must purchase the entire premium at par. The conditions under which the financier absorbs the swapped risk at zero premium are discussed with the calibration in Section 5.6; the producer-side counterpart is developed in Li and Yu (2026).

Bridge to the empirical calibration. The two-state primitives map to the empirical KDE on SAF prices through a median collapse at the bin-8/9 boundary, $\pi_H = 0.472$, $w_H = \$7.16/\text{gal}$, $w_L = \$4.97/\text{gal}$ (Section 5.3). The mapping rests on one substantive assumption: the conditional means are moments of the *realized price*, while the closed form’s $w(\omega)$ is a demand intercept. The bridge is the parallel-shift scenario mapping of Section 3.5, under which a price realization shifts the willingness-to-pay intercept one-for-one, so realized price dispersion is treated as dispersion in the state-conditional net premium $A(\omega) = w(\omega) - c$. The closed form is agnostic between demand- and cost-side sources of that dispersion provided the two do not covary; positive cost–price covariance would compress the spread and lower σ^* (Section 6.1). Table 1 summarizes the comparative statics, and Figures 1 and 2 plot σ^* and EVPI against their primitives at the baseline calibration.

3. Empirical Model

We discretize (3) into a two-stage stochastic mixed-integer linear program, with first-stage facility binaries shared across scenarios.

3.1. Decision variables

- $\text{DvD}_{d,a} \in \{0, 1\}$: depot at county d in ash category a , $d \in \mathcal{D}$, $a \in \mathcal{A}_d$.
- $\text{DvB}_{r,a'} \in \{0, 1\}$: refinery at county r in ash category a' , $r \in \mathcal{R}$, $a' \in \mathcal{A}_r$.

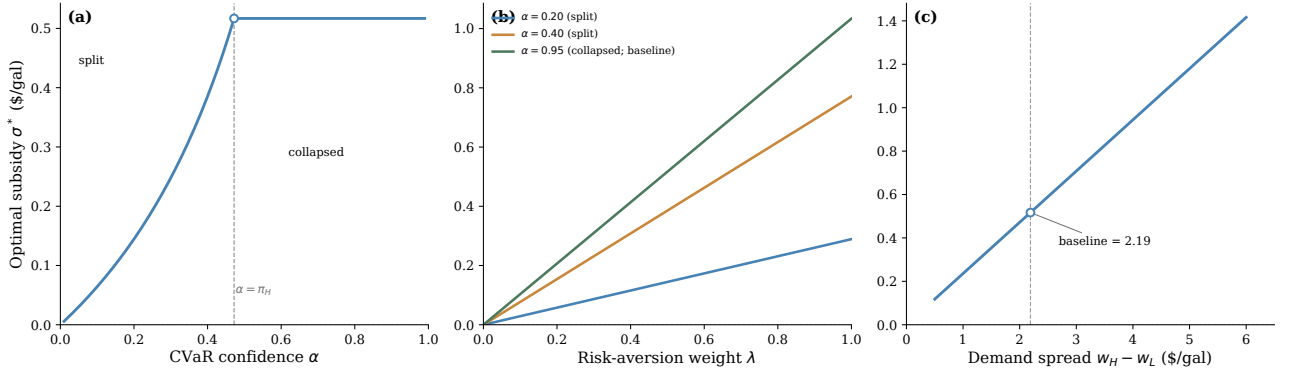


Figure 1: Comparative statics of σ^* . (a) $\sigma^*(\alpha)$, continuous at the regime boundary $\alpha = \pi_H$ with a kink in the derivative (Corollary 3). (b) $\sigma^*(\lambda)$ at three CVaR confidence levels. (c) $\sigma^*(w_H - w_L)$ at the baseline $(\lambda, \alpha, \pi_H) = (0.5, 0.95, 0.472)$, with the KDE-anchored spread \$2.19/gal marked.

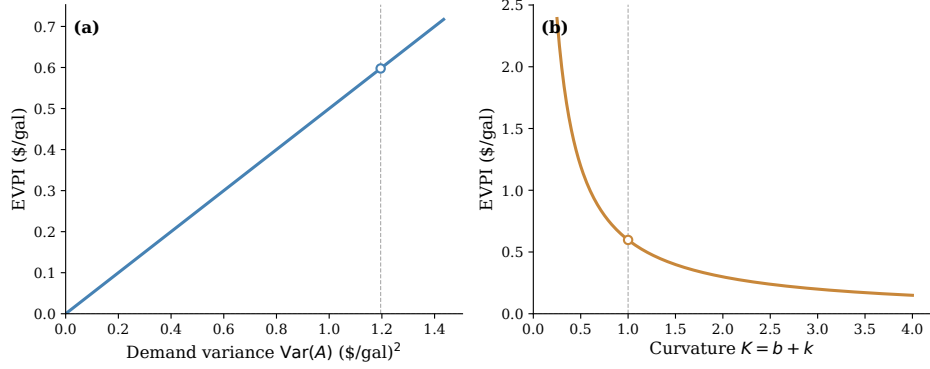


Figure 2: Sensitivity of EVPI to its two primitives. (a) EVPI against $\text{Var}(A)$ via the demand-state spread. (b) EVPI against the combined curvature $K = b + k$. Both panels plot the closed form (9) directly at the baseline calibration.

- $x_{h,d,a}^\omega \geq 0$: residue (dt, dry tons) from h to depot (d, a) in scenario ω .
- $z_{d,a,r,a'}^\omega \geq 0$: pre-processed flow depot $(d, a) \rightarrow$ refinery (r, a') in scenario ω .
- $\text{slack}^\omega \geq 0$: imported SAF in scenario ω .

3.2. Constraints

Supply. For all $h \in \mathcal{H}$, $\omega \in \mathcal{S}$:

$$(13) \quad \sum_{d \in \mathcal{D}} \sum_{a \in \mathcal{A}_d} x_{h,d,a}^\omega \leq \bar{R}_h,$$

where $\bar{R}_h$ is the Stage-1-derived available residue (county aggregation: $\bar{R}_F = \sum_h \theta_{h,F} \cdot \bar{R}_h^{\text{hex}}$, with $\theta_{h,F}$ the hex-to-county area share; see Section B).

Depot capacity & ash bounds. For all (d, a) , ω :

$$(14) \quad \sum_h x_{h,d,a}^\omega \cdot \text{Res2Pre} \geq \text{DvD}_{d,a} \cdot \kappa_{lw}^D,$$

$$\begin{aligned}
(15) \quad & \sum_h x_{h,d,a}^\omega \cdot \text{Res2Pre} \leq \text{DvD}_{d,a} \cdot \kappa_{\text{up}}^D, \\
(16) \quad & \sum_h \text{AshStd}_h \cdot \xi(\omega) \cdot x_{h,d,a}^\omega \geq \text{DvD}_{d,a} \cdot \kappa_{\text{lw}}^D \cdot \pi_a^D, \\
(17) \quad & \sum_h \text{AshStd}_h \cdot \xi(\omega) \cdot x_{h,d,a}^\omega \leq \text{DvD}_{d,a} \cdot \kappa_{\text{up}}^D \cdot \bar{\pi}_a^D, \\
(18) \quad & \sum_a \text{DvD}_{d,a} \leq 1.
\end{aligned}$$

Depot mass balance & refinery capacity / ash. For all (d, a) and (r, a') :

$$\begin{aligned}
(19) \quad & \sum_{r,a'} z_{d,a,r,a'}^\omega = \text{Res2Pre} \sum_h x_{h,d,a}^\omega, \\
(20) \quad & \sum_{d,a} \beta \cdot \phi(a') \cdot z_{d,a,r,a'}^\omega \geq \text{DvB}_{r,a'} \cdot \kappa_{\text{lw}}^R, \\
(21) \quad & \sum_{d,a} \beta \cdot \phi(a') \cdot z_{d,a,r,a'}^\omega \leq \text{DvB}_{r,a'} \cdot \kappa_{\text{up}}^R, \\
(22) \quad & \sum_{a'} \text{DvB}_{r,a'} \leq 1, \\
(23) \quad & \sum_{d,a} \text{DAshMean}_a \cdot z_{d,a,r,a'}^\omega \geq \frac{\kappa_{\text{up}}^R}{\beta} \cdot \text{DvB}_{r,a'} \cdot \{\pi_{a'}^R, \bar{\pi}_{a'}^R\}.
\end{aligned}$$

Here $\text{Res2Pre} = 0.85$ is the depot pretreatment recovery (the fraction of delivered residue that survives pre-processing), AshStd_h is the standardized ash content of county h 's residue, DAshMean_a is the mean ash content of depot ash class a , and the κ and π ($\bar{\pi}$) constants are capacity and ash-share lower (upper) bounds at depots (D) and refineries (R).

Market clearing. For all ω :

$$(24) \quad \text{slack}^\omega + \sum_{d,a,r,a'} \beta \cdot \phi(a') \cdot z_{d,a,r,a'}^\omega \geq \bar{Q},$$

where β denotes the biofuel conversion yield (gal of SAF per dry ton of residue).

3.3. Objective: Discretized social welfare

The discretized form of (3) (assuming horizontal demand, risk-neutral; the endogenous-demand variant used for the regret analysis is specified in Section 3.5, and risk aversion in Section 2.3) is as follows. Under horizontal demand the consumer-surplus integral in (1) is identically zero, so this fixed-demand objective is the chain surplus (producer surplus net of the slack opportunity cost), which is the accounting behind the information-value

check of Section D.

$$\begin{aligned}
 (25) \quad \max \quad & \frac{1}{1+\rho} \sum_{\omega} p^{\omega} \left[p_{\text{SAF}}(\omega) \sum_{d,a,r,a'} \beta \phi(a') z_{d,a,r,a'}^{\omega} - \text{Stump}^{\omega} - \text{Harvest}^{\omega} \right. \\
 & - \text{Transport}_{H \rightarrow D}^{\omega} - \text{Transport}_{D \rightarrow R}^{\omega} - \text{OPEX}^{D,\omega} - \text{OPEX}^{R,\omega} \\
 & \left. - \text{Slack penalty}^{\omega} - p_C \cdot E(\mathbf{x}^{\omega}, \mathbf{z}^{\omega}, y; \omega) \right] - \text{CAPEX}^D - \text{CAPEX}^R,
 \end{aligned}$$

where OPEX denotes operating expenditure and CAPEX denotes capital expenditure. CAPEX terms are first-stage (no ω index, just y -linear); all others scenario-indexed.

3.4. County-level aggregation

The Stage-2 operational solve runs at the county level. Hex quantities aggregate to counties via area-share weighting; an invariance result for the harvest-cost term (Proposition 3, proved in Section B) supports the aggregation, with the remaining objective terms aggregating by summation or by the distance-averaging linearization documented there. The operative bound is empirical: county-level total NPV cost matches the hex-level value to within $\pm 0.15\%$ (Table 4).

3.5. Endogenous-demand variant for the regret analysis

The fixed-demand MILP above prices output at the exogenous $p_{\text{SAF}}(\omega)$ against a hard demand floor $\bar{Q}$ with slack imports; it produces the deterministic baseline of Section 5.1 and the fixed-demand price sweep of Figure 8. The regret statistics of Section 5.2 and the candidate evaluation of Section 5.4 are computed on an *endogenous-demand* variant of the same MILP, in which the demand floor and the slack term are removed and demand clears along a linear inverse-demand curve. The closed-form objects of Section 2.4 are calibrated to the KDE alone and are variant-independent. The variant differs from the fixed-demand model in three respects.

Demand and calibration. Inverse demand is $P^d(Q; \omega) = w_{\omega} - bQ$, with slope pinned by an elasticity at an anchor point via $b = -P^{\star}/(\varepsilon Q^{\star})$ and $w = P^{\star} + bQ^{\star}$. The regret evaluation is calibrated at $\varepsilon = -0.5$, the midpoint of the $\{-0.3, -0.5, -1.0\}$ grid of Section 4.4, and the raw-mean anchor $(Q^{\star}, P^{\star}) = (3.0 \text{ B gal/yr}, \$5.97/\text{gal})$, giving $w = \$17.91/\text{gal}$. Because calibrations differ in welfare scale, the level of $\mathbb{E}[L]$ is calibration-specific; the first-stage topology is invariant across the elasticity grid (Section 4.4).²

Scenario mapping. Price scenarios enter as parallel intercept shifts, $w_{\omega} = w + (p_{\text{SAF}}(\omega) - \$5.977)$, where $\$5.977$ is the raw sample mean (quoted as $\$5.97$ elsewhere in the paper); the $\$0.03$ raw-vs-binned gap shifts all intercepts equally and cancels from every L_s . Holding b fixed, a scenario shifts the buyer-side willingness to pay one-for-one with the wholesale price while preserving the local elasticity at the shifted anchor.

Objective. The revenue term and slack penalty in (25) are replaced by the gross-benefit integral $\int_0^Q P^d(q; \omega) dq = w_{\omega}Q - \frac{b}{2}Q^2$, linearized by piecewise-linear interpolation with type-2 special-ordered sets (SOS2) on a shared 21-breakpoint grid (cross-plan comparisons are re-evaluated at 81 breakpoints;

²At $\varepsilon = -1.0$ the low-price wait-and-see MILPs do not converge within the time budget, so their recorded regret is not evidence either way; Section C documents the residual gaps. Whether the tail-to-mean ratio (CVaR divided by the mean loss) is less calibration-sensitive than the level is an open robustness item.

Section D), so the program remains a MILP; maximizing it recovers CS + PS.

Each scenario’s wait-and-see value is a full MILP solve at a 5% target gap followed by a certification pass against the candidate plans of Section 5.4, and each here-and-now value fixes the first-stage sitings and solves the recourse problem to numerical optimality; the protocol, solve limits, and realized gaps are reported in Sections C and D.

4. Data and Calibration

Four inputs calibrate the program: the spatial frame, residue supply and SAF demand, the SAF price distribution (Section 4.3), and the cost parameters (Section 4.4).

4.1. Spatial frame

The spatial frame is the 14,040 50-sq-mile hex tessellation covering the 12 core SE-U.S. study states (Texas through Maryland) adopted from Li et al. (2026). We aggregate hex-level quantities to 1,374 county-level supply nodes via U.S. Census Bureau (2023) county polygons; 8,864 hexes (63%) straddle multiple counties and are assigned by area-share weighting (Section B). Proposition 3 establishes that the harvest-cost term aggregates without distortion, and the full accounting stack matches across resolutions to within $\pm 0.15\%$ (Table 4). The Stage-2 instance draws residue from the 220 counties with positive Year-0 hardwood availability and sites depots and biorefineries on 483 candidate counties each, those containing at least one of the 528 industrially zoned hexes, with both catchment legs capped at 100 miles; it is solved over a ten-year horizon against the 20 price scenarios of Section 4.3. Table 5 (Section B) collects the instance dimensions and Table 9 (Section D) the solver profile.

4.2. Residue and demand

Annual residue availability $\bar{R}_h$ is the Stage-1 LP solution under mill demand projections from USDA Forest Service (2023a), with willingness-to-harvest $W2HF = 0.28$ (Butler et al. 2016), commercial-private share $ComR = 0.338$, and residue:wood ratios from county-level U.S. Department of Agriculture (USDA) Forest Service Timber Products Output (TPO) removals data (USDA Forest Service 2023b), assembled at hex resolution as in Li et al. (2026). The hardwood pool delivers 72.5 M dt/yr at Year 0 across 220 supply counties, of which the deterministic baseline of Section 5.1 draws 56.7 M dt/yr (78%). The pool is harvest-driven, mill-demand-determined removals times county residue:wood ratios, scaled by the 0.70 retrieval cap the Billion-Ton assessments impose on logging residues (Langholtz et al. 2024, Ch. 4; U.S. Department of Energy 2016); the willingness-to-harvest and commercial-private parameters enter as an inventory cap that is slack at the demand-driven harvest level. It is a technical-availability envelope, not comparable to the 2023 Billion-Ton Report’s production-capacity figures (30.3 M dt/yr nationally across forest resource types, with roughly 19 M dt/yr of logging residues at stumpage prices up to \$40/dt), and the binding economics in the model come from harvest, transport, and processing costs rather than from the envelope. The demand floor is $\bar{Q} = 3.0$ B gal/yr, the 2030 interim target of the U.S. SAF Grand Challenge (U.S. Department of Energy and U.S. Department of Transportation and U.S. Department of Agriculture 2021), read as a long-run equilibrium volume for the chain’s operating window rather than a near-term forecast; instrument outlays in Section 5.6 are quoted per this volume and scale linearly in it. The baseline is hardwood-only; a softwood-inclusive sensitivity (pool 117.0 M dt/yr)

returns the same 248/34 first-stage topology in every cell of the robustness grid (Section 6.2). Catchment-radius construction, the sustainable-removal fraction, and the hex-county aggregation are documented in Section B.

4.3. SAF price uncertainty

Because a deep SAF spot market does not yet exist, the uncertain revenue price is a *constructed drop-in biofuel price*: a slate-weighted wholesale gasoline–diesel price adjusted to a common real basis, plus the market value of the Renewable Fuel Standard (RFS) compliance credits the fuel generates, the weekly D-code Renewable Identification Number (RIN) price scaled by the 1.7 cellulosic equivalence factor. The series spans August 2013 through December 2021 and is the producer-revenue price of the prior fast-pyrolysis/hydrotreatment (FP/HT) pathway calibration (Li et al. 2026); its construction and sources are documented in Section C. Its *level* embeds RIN value (on average \$2.96/gal of the \$5.98/gal mean), so the price is policy-inclusive and the non-capitalization hypothesis of Proposition 1 must be re-argued (Section 5.5). Its *dispersion* is genuine revenue risk, since RIN-price volatility passes one-for-one into the margin of a producer paid partly in credits, and the dispersion is what the risk objects of Section 2.4 price. The outside-option price $p^{\text{jet}} = \$3.50/\text{gal}$ (Section 2.2) is a fossil-jet benchmark on a policy-free basis; the wedge between the two prices is predominantly the embedded RIN component. We discretize the distribution with a Gaussian KDE on 20 equal-width bins (Table 8, Section C). The raw sample mean is \$5.97/gal and the probability-weighted binned mean is $\overline{p}_{\text{SAF}} = \$6.00/\text{gal}$ on support [\$3.81, \$9.12]/gal; discretization-derived quantities use the binned mean, and the deterministic baseline of Section 5.1 is the central-scenario solve at bin 9’s conditional mean \$6.05, the bin containing the raw mean.

4.4. Cost parameters

Feedstock quantities and residue-harvest costs follow U.S. Department of Energy (2016), with year-0 representative values discounted at $\rho = 0.10$; all costs are in 2022 USD, with the CPI factor 1.223 applied to the 2016 residue-harvest-cost values. Conversion-side capacities and unit costs come from two techno-economic analysis (TEA) models. The biorefinery basis is a gasification–Fischer–Tropsch TEA in the lineage of Tanzil et al. (2021): total capital investment (TCI) is built up from delivered equipment cost using the ratio factors of Peters, Timmerhaus, and West (2004) on the reference design and process flows of Swanson et al. (2010), escalated with the Chemical Engineering Plant Cost Index, placed on a 100 M gal/yr anchor, scaled to other capacities with a 0.7 power law, taken net of a 10% salvage value in year 20, and annualized over a 20-year life at $\rho = 0.10$. This yields \$947 M of per-facility capital and an annualized charge of \$111.3 M/yr (Table 11). The depot basis is this project’s own preprocessing TEA, resolved separately for the hammer-mill and crumbler configurations; the crumbler configuration is carried into the results.

Comparison against public design cases. The charge is fixed per facility, so its per-gallon reading falls as realized output rises: the \$947 M spans \$9.5–\$19/gal-yr across the admissible [50, 100] M gal/yr band, and the baseline footprint operates near the upper end: 3.32 B gal/yr over 34 refineries is 97.5 M gal/yr each, or $\approx \$9.7/\text{gal-yr}$. Table 11 (Section D) places that figure against the public National Renewable Energy Laboratory (NREL) design cases: the thermochemical indirect-liquefaction design implies \$9.31/gal-yr of operating output on an *n*th-plant basis (Tan et al. 2015, 2017), and the 2018 biochemical design cases \$22–23 per annual gallon of gasoline equivalent (GGE) (Davis et al. 2018). Our \$9.7 is a $\approx 1.04\times$ premium over the thermochemical

n th-plant anchor, conservative relative to the $\approx 2\times$ pioneer-plant cost growth documented by Merrow, Phillips, and Myers (1981); neither design case enters the optimization.

Operating cost is \$1.96/gal at the baseline, the same TEA’s operating-cost output for this feedstock and product slate: gross operating cost per dry tonne of feedstock, less the credit for the diesel, gasoline, and naphtha co-produced alongside the jet fraction, converted to a per-gallon basis at the pathway’s liquid yield. The figure is therefore net of co-product revenue, which is why it sits well below the gross conversion cost. The operative dry-ash-free yield of 92.84 gal/dt (BCY paragraph below) sits at the hydrogen-boosted end of gasification–FT performance; if achieving it requires purchased hydrogen, \$1.96 understates operating cost, and the high-yield/low-OPEX pairing is an optimistic baseline. The sensitivity grid brackets exactly that possibility.³ Finally, we state what the TEA parameters do and do not bind. They do not bind the headline σ^* , which depends only on the risk-aversion weight and the KDE primitives and is numerically invariant across the grid by construction (Theorem 1). They do bind the build-vs-slack crossing and the regret levels. Re-solving the crossing region at each of the three operating costs shows that every per-gallon dollar of operating cost moves the crossing by slightly more than a dollar — between $1.1\times$ and $1.2\times$ across the two steps, against the one-for-one shift a back-of-envelope argument would give (Table 10). The crossing therefore rises faster than the operating-cost assumption itself, and the number of low-price bins that fall below the build threshold — the bins that carry essentially all of the regret — grows accordingly.

Biofuel conversion yield (BCY). Following the ash-tier discretization method of Li et al. (2026), we discretize BCY into 10 percentile-based tiers spanning 63.8–95.6 gal/dt across the hardwood-residue ash distribution, anchored to a base yield of 75.16 gal/dt (ash-inclusive basis) for the median-ash tier consistent with FT-from-residue performance in Tan et al. (2017); the model’s operative constant is the dry-ash-free equivalent, $75.16/0.810 = 92.84$ gal/dt, to which the per-tier ash scale factors apply. The FT pathway used here has yield characteristics distinct from the FP/HT pathway analyzed in Li et al. (2026); we adopt the ash-tier *discretization scheme* from that work while substituting the FT-base BCY of ~ 75 gal/dt.

5. Results

The deterministic baseline anchors everything that follows. Three conventions apply throughout. First, unless a table row is labeled otherwise, all reported quantities are single-year-equivalent flows (\$/yr in year-0 terms). Second, operations are evaluated over a 10-year decision horizon, with capital annuitized over a 20-year asset life at $\rho = 0.10$. Third, instrument outlays are quoted at the 3.0 B gal/yr Grand Challenge anchor volume, so they do not depend on which deterministic run defines the baseline. The regret statistics of Section 5.2 and the candidate evaluation of Section 5.4 use the endogenous-demand variant (Section 3.5); the closed-form objects of Sections 5.3, 5.5 and 5.6 are calibrated to the KDE alone.

³Two higher operating costs, \$2.30 and \$2.70/gal, price an H₂-inclusive configuration; the grid crosses the three costs with the three demand elasticities of Section 6.1 and is reported in Table 10 (Section D). The 248-depot / 34-refinery first-stage topology holds in eight of the nine cells; the exception is the elastic, highest-cost corner ($\varepsilon = -1.0$, \$2.70/gal), which returns 185/27 at a 10.6% realized gap and so is an unconverged incumbent rather than a certified topology shift (an incumbent is the best solution the solver has found so far; a certified result is proven optimal within the stated gap). Its direction, a smaller footprint under elastic demand and high conversion cost, is nonetheless the economically expected one. The softwood-inclusive re-solve of the same grid (Section 4) returns 248/34 in all nine cells.

Table 2: State-level distribution of placed depots and refineries at the deterministic baseline (KDE-mean scenario, $p_{\text{SAF}} = \$5.97/\text{gal}$)

State	Depots	Share (%)	Refineries	Share (%)
Tennessee (TN)	52	21.0	6	17.6
Virginia (VA)	46	18.5	7	20.6
North Carolina (NC)	34	13.7	4	11.8
Georgia (GA)	32	12.9	4	11.8
Alabama (AL)	26	10.5	4	11.8
Mississippi (MS)	20	8.1	2	5.9
South Carolina (SC)	13	5.2	3	8.8
Arkansas (AR)	13	5.2	2	5.9
Kentucky (KY)	8	3.2	1	2.9
Louisiana (LA)	4	1.6	1	2.9
Total	248	100.0	34	100.0

Notes: The SE-U.S. core (TN, VA, NC, GA, AL) hosts 77% of all depots and 74% of all refineries; remaining placements track the residue resource base across Mississippi, Arkansas, South Carolina, Kentucky, and Louisiana.

5.1. Deterministic baseline

We first solve the deterministic single-scenario problem at the central scenario, bin 9 of Table 8, whose conditional mean $p_{\text{SAF}} = \$6.05/\text{gal}$ is the bin containing the raw sample mean $\$5.97$. The incumbent configuration places $N_D = 248$ depots and $N_R = 34$ biorefineries across 10 of the 12 study states, producing 3.32 B gal/yr of domestic SAF and fully meeting the 3.0 B gal demand floor with zero slack imports; present-value chain surplus (PVC) is $\$1.43$ B/yr.⁴ The siting tracks the SE-U.S. forest-residue base (Figure 3); state-level shares are reported in Table 2. Sweeping the same fixed-demand model across the 20 price bins (Figure 8 in Section F) locates the build-versus-import crossing at roughly $\$5.5/\text{gal}$, where the chain surplus turns positive, and an expansion band over $\$5.8$ – $\$6.6/\text{gal}$ across which the chain adds capacity beyond the baseline siting. The low-price scenarios that carry the regret of the next subsection sit at or below that crossing.

5.2. Price uncertainty and the low-price tail

The welfare cost of committing early is concentrated in the low-price tail. Take the central plan $y_{\text{bin } 9}^*$ as the here-and-now commitment and, for each of the 20 KDE bins of the endogenous-demand variant, define the loss $L_s = W_s^{\text{WS}} - W_s^{\text{HN}}$: the scenario- s wait-and-see value less the realized welfare of the committed plan in that scenario. Figure 4 plots the twenty losses. The loss declines monotonically across the nine lowest-price bins, from $\$947$ M/yr at bin 1 to $\$48$ M/yr at bin 9, and is zero in the 11 highest-price bins (about 41% of the probability mass), where the wait-and-see topology coincides with the committed plan. Expected regret is $\mathbb{E}[L] = \$299$ M/yr and the tail is $\text{CVaR}_{0.95}[L] = \921 M/yr, a tail-to-mean ratio of 3.1.⁵

⁴The bin-9 plan is a feasible incumbent, not a certified optimum; Section D reports the source runs, a companion re-solve at the raw mean price, and their realized gaps. The regret analysis of Section 5.2 treats it as a fixed commitment.

⁵The worst-5% window comprises bin 1 in full plus the lowest 29.0% of bin 2's mass (Table 8). Every wait-and-see value is certified against the candidate pool of Section 5.4, and every here-and-now value is a fixed-siting solve with recorded gap below 10^{-12} . Because

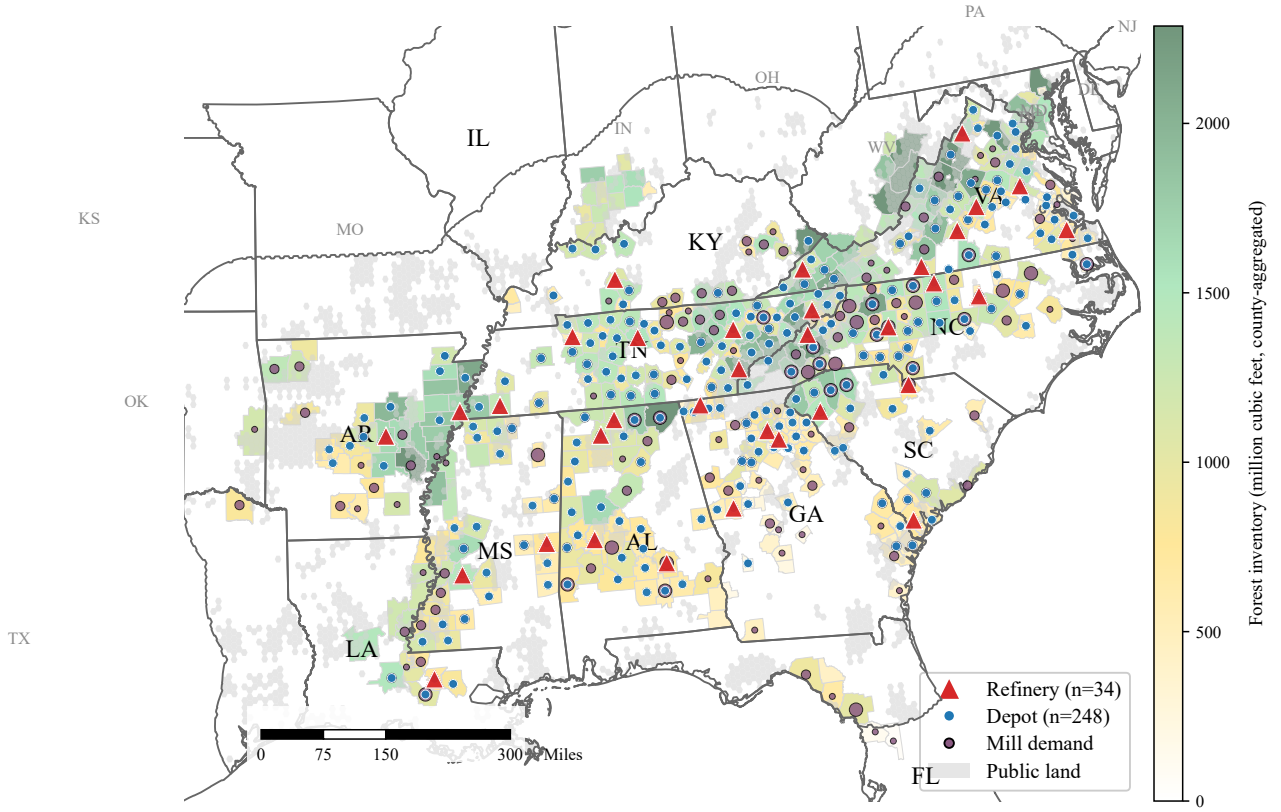


Figure 3: County-level baseline siting map: forest residue inventory (background shading) overlaid with the placed depots and refineries of the deterministic baseline plan (bin-9 central scenario, $p_{SAF} = \$6.05/\text{gal}$; 248 depots, 34 refineries).

Losses concentrate because the committed plan overbuilds; the wait-and-see topologies trace this directly. The scenario-optimal siting is 192/26 depots/refineries at bin 1, 197/27 across bins 2–9, and the committed 248/34 from bin 10 upward, a weakly monotone expansion in price; in bins 1–9 the committed plan’s excess capital relative to the smaller ideals is the regret. At high prices marginal capacity beyond the central scale earns little, because incremental catchment residue is expensive and the consumer-surplus integral flattens, and no wait-and-see topology exceeds 248/34 in any bin. Commitment risk is therefore asymmetric: overbuilding against a low price is costly, while underbuilding against a high price is cheap at the margin.

Regret and the CVaR functional of (4) are different objects that point at the same states. Regret measures the cost of having committed before the price was known. The functional instead prices tail risk in welfare levels, and the welfare-level tail generates σ^* in Theorem 1. The two coincide in location: CVaR_{0.95} of the committed plan’s welfare level sits 33% below its expectation, carried by the two lowest-price bins, so the states that carry the regret are the states the mean–CVaR objective weights. This motivates the objective in (4) and the risk-premium subsidy of the next subsection.

the wait-and-see values are converged incumbents (gaps 1.3–3.9%), the figures are lower bounds on true regret; the gap-implied allowance (the range regret could take given the solver’s residual gap) bounds $E[L]$ within $[\$299, \$827]$ M/yr and the tail-to-mean ratio within $[1.1, 4.1]$ (Table 7; Sections C and D). The same information question, asked on the fixed-demand chain-surplus accounting, gives a candidate-based EVPI estimate of $\$255$ M/yr, the same order as the regret (Table 13, Section D).

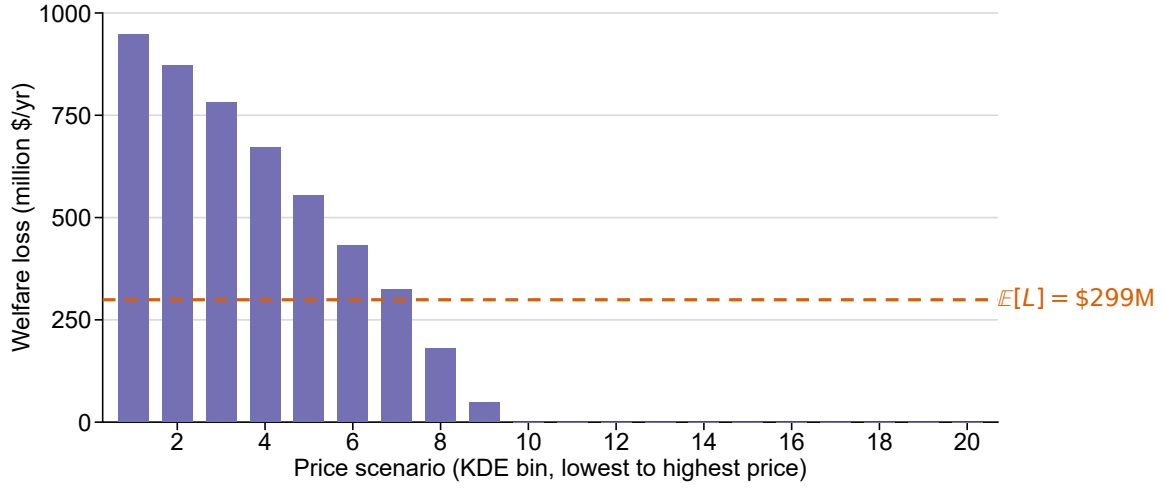


Figure 4: Per-scenario welfare loss $L_s = W_s^{\text{WS}} - W_s^{\text{HN}}$ of the committed here-and-now plan $y_{\text{bin}9}^*$ across the 20 KDE-discretized SAF-price scenarios (endogenous-demand variant, Section 3.5; realized wait-and-see gaps 1.3–3.9%). The loss declines monotonically from \$947 M/yr at bin 1 to \$48 M/yr at bin 9, where the wait-and-see ideal is a structurally smaller chain (192/26 at bin 1, 197/27 at bins 2–9), and is zero from bin 10 upward. The dashed line marks $\mathbb{E}[L] = \$299 \text{ M/yr}$; $\text{CVaR}_{0.95}[L] = \921 M/yr is carried by the two lowest-price bins. Losses are lower bounds on true regret up to the residual gaps.

5.3. A welfare-equivalent subsidy below the 45Z credit

The risk-premium subsidy σ^* is the per-gallon transfer that restores risk-neutral capacity for a mean–CVaR planner facing price risk and irreversible investment. It prices the low-price welfare tail of Section 5.2 but is calibrated to the KDE primitives directly rather than to either MILP variant. In the collapsed regime $\alpha \geq \pi_H$ of Theorem 1 (the case where CVaR collapses to the low-price state’s welfare level rather than averaging over part of it), which holds at $\alpha = 0.95$ and the KDE-implied $\pi_H = 0.472$,

$$(26) \quad \sigma^* = \lambda \pi_H (w_H - w_L).$$

The 20-bin KDE (Table 8) admits a two-state collapse (the 20 price bins reduce to two effective states) at the bin-8/9 boundary (\$5.93, the bin edge nearest the binned mean of \$6.00/gal): $\pi_H = 0.472$ above and $\pi_L = 0.528$ below, with conditional means $w_H = \$7.16/\text{gal}$ and $w_L = \$4.97/\text{gal}$, so the state-conditional spread is $w_H - w_L = \$2.19/\text{gal}$. At the midpoint risk weight $\lambda = 0.5$, (26) gives $\sigma^* = 0.5 \times 0.472 \times \$2.19 \approx \$0.52/\text{gal}$.

We report σ^* as a schedule in λ rather than as a point. At fixed KDE primitives the closed form is linear, $\sigma^*(\lambda) = \lambda \times \$1.04/\text{gal}$, so the admissible range $\lambda \in [0, 1]$ maps to $\sigma^* \in [0, \$1.04]/\text{gal}$. The endpoint $\lambda = 0$ is the risk-neutral planner of Weitzman (1974), with $\sigma^* = 0$. The endpoint $\lambda = 1$ is the tail-only planner. Its closed-form optimum is the deterministic optimum at the low-state conditional mean, and its nearest empirical counterpart is the bin-1 wait-and-see plan of 192 depots and 26 refineries. One statement holds at every risk weight: since $\sigma^*(\lambda) \leq \$1.04 < \$1.30/\text{gal}$, *price risk alone cannot cover the statutory credit*. Where a point is needed we use $\lambda = 0.5$ and carry λ explicitly in every downstream object.⁶

Two things follow. First, $\sigma^* = \$0.52/\text{gal}$ sits materially below the statutory \$1.30–\$1.45/gal credit, and the

⁶An external anchor from observed project-finance hurdle premia, inverting the closed form through the capital stack, belongs to the companion paper’s project-finance setting (Li and Yu 2026).

shortfall is not an artifact of the midpoint: even $\lambda = 1$ delivers only \$1.04/gal. Section 5.5 asks how much of the remainder the carbon externality accounts for. Second, the closed form contains neither marginal cost nor curvature, so cost-level feedbacks leave σ^* unchanged. Channels that act on the committing agent’s effective risk aversion move it instead: financing frictions and hurdle-rate wedges raise σ^* , while fixed-price offtake coverage, positive co-movement of input costs with fuel prices, and partial financial hedging lower it. Both signs are carried through Section 6.1.

5.4. Validating the risk premium on the integer chain

The closed form prices a contraction: a tail-averse planner builds less than a risk-neutral one, and σ^* is the transfer that undoes the difference. Two evaluations on the MILP check that the integer chain behaves the same way.

The first holds each of six candidate first-stage plans fixed — the committed 248/34 and the wait-and-see topologies of the low-price bins, 192/26, 195/26, 197/27, 204/28, and 211/30 (Table 6, Section C) — and evaluates it by fixed-siting recourse solves across all 20 bins on an 81-breakpoint grid (Section D). Expectation and regret on one side, and the welfare-level tail on the other, rank the plans in opposite directions. The committed plan wins on every regret-based measure. Its expected welfare is $\mathbb{E}[W] = \$17.004$ against the 197/27 plan’s \$16.904 B/yr. Its expected loss is $\mathbb{E}[L] = \$300$ M/yr against \$400 M/yr, and its tail loss is $\text{CVaR}_{0.95}[L] = \931 M/yr against \$1,837 M/yr. These are the committed plan’s own statistics under this subsection’s grid refinement, moving from the \$299/\$921 M of Figure 4 to \$300/\$931 M. The 197/27 plan, in turn, carries the welfare-level tail: its $\text{CVaR}_{0.95}[W]$ is higher by \$0.93 B/yr. The mean–CVaR preference therefore switches at $\lambda^* \approx 0.10$; above it the contracted plan is preferred, by \$0.41 B/yr at $\lambda = 0.5$. The disagreement is by construction: regret charges the smaller plan for the upside it forgoes at high prices, whereas welfare levels face the tail where it lives, and there the smaller plan’s avoided capital is a pure gain. Regret is the right lens for the commitment question. The welfare-level object governs the instrument question instead: what a tail-averse planner builds, and what a subsidy must restore. That the tail-preferred plan is a one-fifth contraction of the committed one ($197/248 = 0.79$ on depots, $27/34 = 0.79$ on refineries) is the model-scale counterpart of the contraction Theorem 1 prices.⁷

The second evaluation injects the credit into the planner’s program and asks whether it undoes the risk distortion. The full 20-bin mean–CVaR extensive form is beyond reach at this instance size, so the test runs on the model’s own two-state collapse with the sitings fixed, so that LP evaluation resolves the ranking to machine precision. At $\lambda = 0$ the ordering is $197/27 \succ 248/34 \succ 192/26$, with the committed plan \$30.8 M/yr behind the leader. At $\lambda = 0.5$ without the credit, risk aversion demotes the committed plan to last and widens its wedge to \$306.9 M/yr. Adding $\sigma^* = \$0.518/\text{gal}$ restores the risk-neutral ordering exactly and closes the wedge to \$47.8 M/yr: the credit undoes 94% of the risk-induced distortion in the plan comparison. The test speaks to the λ - and credit-comparative statics that Theorem 1 asserts, not to plan levels (Section D). The MILP also disciplines the closed form’s two free cost parameters. Regressing the per-bin demand intercept on realized wait-and-see output along the first-order condition $w_\omega = c + K Q_\omega$ gives $K = 4.3 \times 10^{-9}$ \$/gal² and $c = \$5.00/\text{gal}$, and predicts a bin-1 contraction of 0.84 against the realized 0.77. The implied closed-form EVPI

⁷The committed plan is an expected-value plan in the sense of (5) up to the \$0.08/gal difference between the bin-9 conditional mean and the distribution mean. Because it is also the pool’s expected-welfare argmax, the candidate-based value of the stochastic solution on this accounting is zero; what tail aversion prices is the \$0.93 B/yr tail shortfall. The joint mean–CVaR program itself is not solved (Section 6.3).

is $\text{Var}(A)/(2K) = \$190 \text{ M/yr}$, the same order as the $\$299 \text{ M/yr}$ regret (Section D).

5.5. Decomposing the 45Z credit: carbon plus risk

Section 45Z is by statute a carbon-intensity-scaled production credit, so its welfare accounting has two components: a Pigouvian payment for avoided emissions and compensation for price risk under irreversible investment.⁸ Proposition 1 establishes that the two are exactly additive, $\sigma_{\text{tot}}^* = \nu + \sigma^*$, so the Pigouvian component can be priced without re-solving the stochastic program.

Under the make-or-buy structure of Section 2.2, every marginal domestic gallon displaces one imported fossil gallon, so the externality value per gallon is the life-cycle emissions differential times the SCC: $\nu = \Delta e \cdot p_C$ with $\Delta e = 10.2 \text{ kgCO}_2\text{e/gal}$ (Section 2.2; certified values under 45ZCF-GREET could differ, Section E). The two federal SCC vintages bracket the price: at the IWG interim value of $\$51/\text{tCO}_2$ (Interagency Working Group on Social Cost of Greenhouse Gases 2021), $\nu = \$0.52/\text{gal}$; at the EPA value of $\$190/\text{tCO}_2$ (U.S. Environmental Protection Agency 2023), $\nu = \$1.94/\text{gal}$. One qualification governs the inversion below. The hypothesis of Proposition 1 that ν is not already reflected in the market intercept does not hold cleanly, because the revenue price embeds RFS compliance value averaging $\$2.96/\text{gal}$ (Section C). The baseline reading is therefore *incremental*: it prices the carbon channel that Section 45Z adds on top of the existing RIN stack, holding that stack in place; the full-capitalization outer bound is reported in Table 3.

Add the risk component $\sigma^* = \$0.52/\text{gal}$, and the two-channel total spans $[\$1.04, \$2.46]/\text{gal}$ across the two vintages, a band that contains the statutory $\$1.30\text{--}\$1.45/\text{gal}$ credit. Invert (10) at $\lambda = 0.5$: the statutory credit is then consistent with an implied SCC of

$$(27) \quad p_C^{\text{implied}} = \frac{\text{credit} - \sigma^*}{\Delta e} \in [\$77, \$91] / \text{tCO}_2,$$

between the two federal vintages (at the unrounded $\sigma^* = \$0.518$). Because the implied SCC scales as $1/\Delta e$, a $\pm 20\%$ band on the life-cycle differential moves it to $\$64\text{--}\76 and $\$96\text{--}\114 , inside the inter-vintage gap in every case.

Equation (27) is one cell of a schedule in four choices: the instrument that delivers the risk component, the statutory vintage, the emissions basis, and the capitalization treatment of the embedded RIN value. Table 3 reports the named readings, and three carry the interpretation. The *instrument-efficient* reading takes Proposition 2 seriously: if the risk component is delivered through the floor/CfD family, the flat credit prices carbon alone and the implied SCC is $\text{credit}/\Delta e = \$127\text{--}\$142/\text{tCO}_2$, the same value that arises at $\lambda = 0$ and under full offtake coverage. The *post-2025* reading prices the credit actually in force: the 2025 reconciliation act aligned the SAF base rate to $\$1.00/\text{gal}$, giving a credit of $\$0.74\text{--}\$0.83/\text{gal}$ and an implied SCC of $\$22\text{--}\$30/\text{tCO}_2$, below the IWG vintage, so the in-force credit is under-sized relative to carbon plus risk at either federal SCC. The *statutory-basis* reading replaces the physical differential with the statute's own $(50 - \text{CI})$ reference, which credits only $37\text{--}41 \text{ kgCO}_2\text{e/mmBTU}$ of avoided emissions against a physical fossil baseline of 93.9 , and raises the implied SCC to $\$169\text{--}\$180/\text{tCO}_2$; we report the physical basis because the welfare-relevant object is physical avoided emissions (Figure 7 in Section F). The between-vintages property therefore belongs to the baseline

⁸The 2025 budget reconciliation act extended 45Z through 2029, aligned the SAF base rate with the non-SAF rate after 2025, and restricted qualifying feedstocks to North America (immaterial here). The baseline targets the as-enacted IRA SAF vintage; the post-2025 rate is repriced in Table 3.

Table 3: Implied-SCC readings of the Section 45Z credit

Reading	Assumption changed	\$/tCO ₂
Baseline (incremental)	as-enacted rate; flat credit; $\lambda = 0.5$; physical Δe	77–91
$\Delta e \pm 20\%$	physical-differential band	64–114
Instrument-efficient	risk via floor/CfD ($= \lambda = 0$; $= \eta = 1$)	127–142
Maximal risk aversion	$\lambda = 1$	26–41
Statutory emissions basis	(50 – CI) replaces physical Δe	169–180
Post-2025 rate	\$1.00/gal base $\Rightarrow$ credit \$0.74–0.83/gal	22–30
Statutory basis + stacking	joint, with \$0.30/gal stacked value (Section 6.1)	≈ 234 –239
Full RIN capitalization	embedded \$2.96/gal read as carbon support	≈ 367 –382

Notes: Each row recomputes (27) under one named change, holding the others at baseline (unrounded $\sigma^* = \$0.518$, $\Delta e = 10.2$ kgCO₂e/gal). The envelope across admissible combinations spans roughly \$22 to \$382/tCO₂: the between-vintages property belongs to the baseline incremental reading, not to the schedule.

incremental reading, and the table shows which departures break it.

The *rationalization frontier* of Figure 5 makes the trade-off explicit: it is the locus of (λ, p_C) pairs consistent with the statutory credit, $0.0102 \cdot p_C + 1.036 \lambda \in [1.30, 1.45]$. A risk-neutral planner rationalizes the credit only at $p_C \in [\$127, \$142]$. At the EPA value the carbon channel alone is \$1.94/gal, exceeding the statutory credit before risk enters. The IWG value rationalizes the credit only at substantial risk aversion, $\lambda \in [0.75, 0.90]$. The question “is 45Z too generous?” has no vintage-independent answer. The decomposition replaces it with a sharper one: the statutory level is consistent with carbon plus risk for any SCC between the two federal vintages at moderate-to-high risk aversion. Because the statutory level is itself CI-scaled, the implied SCC is not a revealed-preference estimate of the drafters’ carbon valuation but the internal-consistency price at which the statutory formula and the risk premium reconcile within one welfare accounting.

5.6. Instrument form: the fiscal cost of paying for risk with a flat credit

The statutory credit pays a flat per-gallon amount in every price state. Proposition 2 shows this is the most expensive way to purchase the risk correction: a flat credit shifts the mean of the payoff distribution, while the distortion σ^* corrects is a variance phenomenon. The committing agent under-builds not because expected returns are too low but because the low-price tail is too costly to bear. A flat credit pays the same amount in the high-price states, where no correction is needed, as in the low-price states that drive the distortion; a price floor concentrates its payments in the low-price states alone; and a two-sided contract removes the variance itself, paying in bad states and collecting in good ones. Calibrating the proposition to the KDE primitives prices the difference.

Floor and CfD. The floor strike that restores risk-neutral capacity, (11), is $\kappa^* = \$5.64/\text{gal}$ at the baseline $(\lambda, \pi_H, w_L, w_H) = (0.5, 0.472, \$4.97, \$7.16)$, six percent below the binned KDE mean of \$6.00.⁹ Its expected outlay (12) is $T_{\text{floor}} = \sigma^* \times \pi_L / (\pi_L + \lambda \pi_H) = \$0.52 \times 0.691 \approx \$0.36/\text{gal}$, a 31% saving on the flat credit; evaluating the same floor directly on the full 20-bin KDE gives $E[(\kappa^* - p_{\text{SAF}})_+] = \$0.37/\text{gal}$, within 4% of

⁹Proposition 2 writes the floor as a contract on the demand intercept; because scenarios shift the intercept one-for-one with the price (Section 3.5), the strike converts to a price floor at the same dollar value.

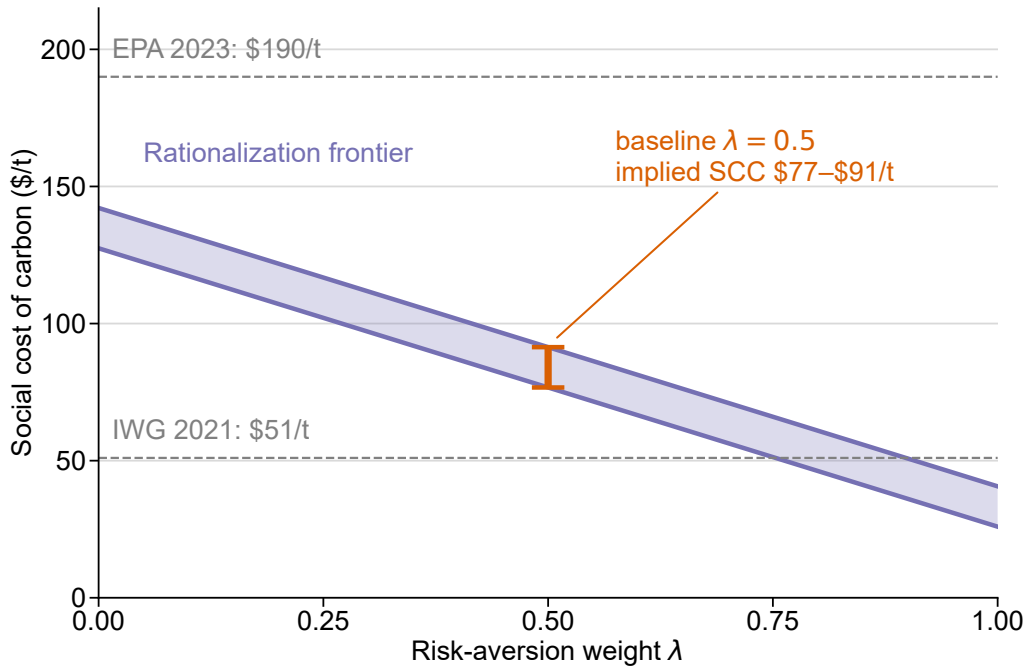


Figure 5: The 45Z rationalization frontier. Shaded band: the locus of risk-aversion weights λ and social costs of carbon p_C at which the two-channel decomposition $\sigma_{\text{tot}}^* = \Delta e \cdot p_C + \lambda \pi_H (w_H - w_L)$ of Proposition 1 exactly equals the statutory \$1.30–\$1.45/gal Section 45Z credit, at the KDE-anchored calibration ($\Delta e = 10.2 \text{ kgCO}_2\text{e/gal}$, $\pi_H(w_H - w_L) = \$1.04/\text{gal}$). Horizontal lines: the two federal SCC vintages (\$51/tCO₂, [Interagency Working Group on Social Cost of Greenhouse Gases 2021](#); \$190/tCO₂, [U.S. Environmental Protection Agency 2023](#)). Marker: the baseline $\lambda = 0.5$, at which the implied SCC is \$77–\$91/tCO₂.

the closed form. At the 3.0 B gal/yr anchor volume the flat credit’s expected outlay for the risk component is \$1.55 B/yr against \$1.07 B/yr for the floor, a saving of \$0.48 B/yr for the identical output correction. A two-sided contract-for-difference at strike equal to the KDE mean eliminates the committing agent’s price variance and restores risk-neutral capacity at *zero* expected outlay (Proposition 2(iii)); the government’s side is actuarially fair but not riskless, absorbing a symmetric exposure with a per-gallon standard deviation of \$1.28.

Whose risk aversion. The result requires the two sides of the swap to price the same risk differently, which Assumption 9 makes coherent. The weight λ prices the exposure of the committing side, the Southeast regional aggregate or the non-diversified investor whose balance sheet carries the irreversible capacity; the federal side pools the swapped exposure across the national taxpayer base, the [Arrow and Lind \(1970\)](#) condition under which its premium tends to zero. Two qualifications apply. Jet-fuel prices are partially correlated with aggregate conditions, which Arrow–Lind does not cover, so the federal side bears a nonzero premium and $T_{\text{CfD}} = 0$ understates the CfD’s social cost; and Corollary 4 maintains a state-independent marginal cost of public funds, so a countercyclical one would add a covariance term.

Strike risk and implementation. The zero-outlay result holds at a strike equal to the true mean, exactly the object a thin SAF market makes hard to observe. A two-sided CfD at strike $\bar{w} + \delta$ has expected outlay δ per gallon, so its advantage over the floor vanishes at a strike error of \$0.36 (6% of the mean) and over the flat credit at \$0.52 (9%); the floor’s outlay rises at slope $\pi_L \approx 0.53$ per dollar of strike error rather than one-for-one. No

deep SAF price index exists, so a contract written today would settle against jet-A plus a premium and leave premium basis risk with the producer. Both considerations leave the floor as the implementable member of the family, on the template of the U.K. Contracts-for-Difference scheme, with auction-discovered strikes, a tenor matched to project debt, and a creditworthy counterparty outside Treasury’s credit machinery (Low Carbon Contracts Company 2025; UK Department for Energy Security and Net Zero 2024; U.S. Department of Energy, Loan Programs Office 2025). The \$0.48 B/yr saving is gross of administrative and transaction costs; Section E collects the institutional detail.

Welfare stakes under costly public funds. At a marginal cost of public funds (MCPF) $\mu = 1$ the instruments are welfare-equivalent and the comparison concerns outlay only (Corollary 4). With the curvature K calibrated from the MILP (Section 5.4), the comparison becomes a benefit–cost test, and it is one-sided. The deadweight loss the risk correction recovers is $\sigma^{*2}/(2K) \approx \31 M/yr (\$31–\$65 M across the K fits), while the cost of paying for it with a flat credit at the empirical federal range $\mu \in [1.10, 1.30]$ (Dahlby 2008) is \$155–\$466 M/yr: the flat credit fails benefit–cost for the risk component at any $\mu > 1.02$. By the same corollary, the floor’s \$0.48 B/yr outlay saving over the flat credit is worth $(\mu - 1) \times \$0.48 \text{ B} \approx \0.14 B/yr in avoided deadweight loss at $\mu = 1.30$. The floor’s own MCPF cost (\$107–\$322 M/yr) also exceeds the recovered loss except at the most favorable corner; only the two-sided CfD passes unambiguously. Costly public funds therefore imply the risk component should not be delivered through an outlay-bearing instrument at all.

Design implication. The decomposition and the ranking together imply a mixed design: the Pigouvian component is a payment for a state-independent externality and is correctly delivered flat, while the risk component is a payment for variance and is more cheaply delivered as a price floor or CfD, the instrument family used in U.K. renewables auctions and in the E.U.’s emerging SAF support schemes. Re-packaging only the risk component of the statutory credit as a floor would hold planner output fixed and save roughly \$0.5 B/yr in expected outlay; the producer-side counterpart, including the project-finance channel through which a floor also relaxes a leveraged developer’s participation constraint, is established in Li and Yu (2026).

6. Discussion and Robustness

This section signs the channels the partial-equilibrium scope omits (Section 6.1), reports the remaining sensitivities (Section 6.2), and states the modeling limitations that bound the results (Section 6.3).

6.1. Partial-equilibrium scope

The welfare functional in (3) excludes four general-equilibrium channels the biofuel welfare literature has found to be material: stumpage-price endogeneity in response to bioenergy demand (Hertel et al. 2010), since we hold stumpage exogenous and treat residue as a by-product of mill demand (USDA Forest Service 2023a); carbon-intensity-based market segmentation and low-carbon fuel standard (LCFS) rents (Holland, Hughes, and Knittel 2009; Khanna, Rajagopal, and Zilberman 2021); partial RIN pass-through under the Renewable Fuel Standard (Lade and Bushnell 2019); and indirect land-use change (Searchinger et al. 2008). The decomposition of Section 5.5 clarifies which component each one moves.

Stumpage endogeneity operates through two routes. On the cost route, an endogenous stumpage response raises the chain’s marginal cost at scale, which moves the level objects but not the risk component, whose closed form contains no cost term; to the extent stumpage prices co-move with fuel prices, the effect on σ^* is downward, because positive covariance between cost and price compresses the spread of the net premium $A(\omega) = w(\omega) - c(\omega)$ that the closed form prices. On the emissions route, stumpage-driven land-use carbon debt erodes Δe ; the oft-cited 25–50 gCO₂e/MJ magnitude is a dedicated-plantation number, and for harvest residues, a by-product of sawtimber-driven rotations, the marginal debt is an order of magnitude smaller. Were the plantation-scale upper end to apply, Δe would fall by 3.3–6.6 kgCO₂e/gal and the between-the-vintages reading would fail; the residue-specific reading keeps the erosion inside the $\pm 20\%$ band of Section 5.5, and indirect land-use change is shut off for the same reason. Segmentation and pass-through operate on the carbon component’s incidence: LCFS credit values ($\sim \$0.20$ – $\$0.35$ /gal) and RIN value passed through at 50–70% are existing partial payments for the same avoided carbon, and where a producer collects them on top of 45Z the SCC required to rationalize the full stack rises, at $\$0.30$ /gal of stacked value from $\$77$ – $\$91$ to approximately $\$106$ – $\$121$ /tCO₂. The risk component is unaffected either way (Proposition 1(iii)). Two channels the decomposition itself omits belong here as well: the private-side participation gap, an incidence question treated in the companion paper (Li and Yu 2026), whose addition to (10) would double count; and the siting response to a carbon price, the marginal-abatement-cost schedule from Pigouvian re-solves of the MILP, left to future work.

Demand elasticity. The closed form and the endogenous-demand variant use linear demand; the fixed-demand model is the $b \rightarrow 0^+$ limit, retained as a computational simplification for the deterministic siting. The headline risk objects are elasticity-invariant by construction, since σ^* and the instrument ranking depend only on the KDE primitives and λ , and the first-stage topology is unchanged across the grid (Section 4.4); the elasticity choice moves only the level of the regret statistics. No direct estimate of a long-run SAF demand elasticity exists, so the grid $\{-0.3, -0.5, -1.0\}$ brackets rather than estimates: the elastic edge is loosely anchored on household flex-fuel switching in Brazil (Salvo and Huse 2013), at best an upper bound for a decade-horizon capacity decision, and the inelastic edge reflects the long-tenor offtake contracts that dominate the U.S. SAF buyer side. Short-run jet-fuel demand under contract is far more inelastic ($\varepsilon \sim -0.15$; Hofer, Dresner, and Windle 2010), but a ten-year siting decision responds to the long-run schedule at which buyers are willing to contract. The regret evaluation uses the midpoint -0.5 .

Offtake coverage. The same long-tenor contracts bound the risk component from *above*. If a share η of output is sold under fixed-price offtake, the committing agent’s effective spread is $(1 - \eta)(w_H - w_L)$ and $\sigma^*(\eta) = (1 - \eta) \sigma^*$. At $\eta = 0.5$, $\sigma^* = \$0.26$ /gal and the implied-SCC band rises to $\$102$ – $\$117$ /tCO₂, still between the federal vintages. We report $\eta = 0$ because contract coverage in a scaled 3 B gal/yr market is unobservable today, but the direction of the bias matters: σ^* is a *lower* bound in the financing-friction and hurdle-wedge channels, which act on λ , and an *upper* bound in offtake coverage, cost–price co-movement, and partial financial hedging. Even at full coverage ($\eta = 1$, $\sigma^* = 0$) the implied-SCC band is credit/ $\Delta e = \$127$ – $\$142$ /tCO₂, so the between-the-vintages reading survives *any* degree of offtake coverage.

Outside-option price. The slack-import price enters the objective only through unmet demand, so the zero-slack baseline has a p^{jet} -invariant surplus. What p^{jet} moves is the build-versus-slack margin, one-for-one: lowering p^{jet}

from \$3.50 to the 2025 range (\$2.10–\$2.60) raises the revenue price at which marginal capacity breaks even by \$0.90–\$1.40/gal. Warm-started re-solves (re-solving from a previous solution rather than from scratch) of the crossing-region bins at $p^{\text{jet}} \in \{2.10, 2.60\}$ confirm that the full-build incumbent dominates the all-import plan at every tested price, by \$4.4 B to \$9.8 B/yr of PVC; whether intermediate topologies dominate full build at the lowest-price bins is not resolved at the realized gaps.

6.2. Other sensitivities

Ash quality operates on the cost and yield side, so by the cost-invariance of the closed form it does not move σ^* ; the 10-tier ash-yield discretization spans a $\pm\$0.30/\text{gal}$ welfare-equivalent SAF-cost range, an order of magnitude smaller than the \$2.19/gal demand-state spread that drives the risk results. County-level aggregation is validated against the underlying 50 mi² hex resolution: aggregate NPV agrees to +0.15%, total residue to –0.07%, and total stumpage cost exactly (Table 4, Section B). A softwood-inclusive re-solve of the nine-cell elasticity $\times$ operating-cost grid returns the 248/34 first-stage topology in every cell (Section 4).

6.3. Limitations

Five modeling simplifications bound the scope.

- (i) *Single-period investment timing.* Depot and refinery decisions are made once at $t = 0$ and held fixed across the 10-year horizon, which overstates first-period capital exposure relative to a staged buildout; the real-options deferral value is bounded above by the value of perfect information (Boomsma, Meade, and Fleten 2012), an object our candidate-based estimate approximates but does not certify (Section D).
- (ii) *Exogenous mill demand.* Residue supply curves and pulpwood demand in fact co-vary; a Faustmann-style supply-side response is left to future work.
- (iii) *Asymmetric substitution structure.* Perfect SAF/fossil-jet substitution is assumed only on the slack-import side; the revenue side treats SAF as a distinct, contract-driven market on the constructed price basis of Section 4.3. The two fuels are substitutes for the buyer of last resort but not for the offtaker buying under an environmental, social, and governance (ESG) mandate, and a constant-elasticity-of-substitution (CES) nest would introduce a substitution elasticity the calibration cannot identify.
- (iv) *Risk-averse first stage not optimized.* The mean–CVaR program (4) is evaluated over a fixed six-plan candidate set (Section 5.4), not solved as a joint MILP: the λ perturbation’s welfare effect sits below the free MILP’s attainable tolerance. Whether the joint optimum at intermediate λ tracks the linear $\sigma^*(\lambda)$ schedule of Theorem 1 or exhibits integer kinks at depot-count transitions remains open, as do converged low-price wait-and-see solves at the $\varepsilon = -1.0$ calibration (Section 3.5).
- (v) *Price-state persistence.* The scenario structure treats the realized price as a persistent decade-scale state. Under decade-scale mean reversion the committed chain earns a multi-year average price whose dispersion is much smaller, and the risk component falls from \$0.52 toward \$0.11/gal (block-bootstrap evidence in Section C); the sample is too short to pin the long-run persistence of the fuel-plus-RIN price level, so this is the primary calibration uncertainty on the risk side.

7. Conclusion

We have decomposed the welfare case for the Section 45Z SAF credit into two statute-relevant pieces, the carbon externality and price risk under irreversible investment, in a two-stage stochastic welfare-maximization model of the SE-U.S. forest-residue supply chain, and asked whether a flat per-gallon credit is the right instrument for the risk component. The closed-form benchmark Theorem 1 maps the empirical price distribution into a risk-premium subsidy of $\sigma^* = \$0.52/\text{gal}$ at the baseline risk weight and at most $\$1.04/\text{gal}$ at any admissible weight, below the statutory $\$1.30\text{--}\$1.45/\text{gal}$ at every λ ; the integer chain reproduces the contraction the closed form prices, and injecting the credit undoes 94% of the risk-induced distortion.

On the credit's *level*, the decomposition is exactly additive (Proposition 1), and carbon plus risk together rationalize the statutory credit at an implied social cost of carbon of $\$77\text{--}\$91/\text{tCO}_2$ under the baseline reading, between the two federal SCC vintages. The implied SCC is a schedule rather than a number (Table 3): reasonable choices of instrument form, statutory vintage, emissions basis, and RIN capitalization move it from $\$22$ to roughly $\$382/\text{tCO}_2$, and the post-2025 rate actually in force prices carbon below the IWG floor. The question “is 45Z too generous?” has no vintage-independent answer: at the EPA (2023) SCC the carbon channel alone exceeds the statutory credit, while at the IWG (2021) value the credit requires substantial risk aversion ($\lambda \approx 0.75\text{--}0.90$) to justify. On the *form*, the risk component is a payment for variance, and a flat credit purchases it at par in every state; a one-sided price floor delivers the identical capacity correction at 69% of the flat credit's expected outlay and a two-sided contract-for-difference at zero (Proposition 2), and with costly public funds the outlay gap is a strict welfare wedge (Corollary 4). The design implication is a mixed instrument: pay the Pigouvian component flat, and pay the risk component through the floor/CfD family already standard in European renewables procurement.

The analysis is partial equilibrium, and the omitted channels move the risk component in both directions (Section 6.1); even at full offtake coverage the implied-SCC band stays between the federal vintages. Three extensions follow: the joint mean-CVaR MILP at intermediate λ , which would test whether the integer chain tracks the linear $\sigma^*(\lambda)$ schedule; a multi-period real-options specification (Dixit and Pindyck 1994; Boomsma, Meade, and Fleten 2012; Insley 2017) that captures the deferral value the present framework bounds above by the value of information; and an endogenous stumpage response (Faustmann 1849; Amacher, Ollikainen, and Koskela 2009) that brings general-equilibrium feedbacks into the welfare object. The producer-side counterpart of the instrument-form result, including explicit 45Z incidence (Lohawala, Toman, and Joiner 2024), is developed in Li and Yu (2026).

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A. Proofs of the closed-form benchmark

This appendix collects the proofs of Lemma 1, Theorem 1, Corollaries 2 to 4, and Propositions 1 and 2.

Throughout, define the net demand premium $A \equiv w(\omega) - c$ with realizations $A_H \equiv w_H - c$ and $A_L \equiv w_L - c$. By Assumption 5, $A_L > 0$; by Assumption 4, $A_H > A_L$. The combined curvature constant is $K \equiv b + k > 0$ under Assumptions 2 and 3.

A.1. Lemma 1 (strict concavity + interior optimum)

PROOF OF LEMMA 1. *Step 1: Strict concavity.* The per-state social welfare under Assumptions 2 and 3 is

$$(28) \quad W(Q; \omega) = (w(\omega) - c)Q - \frac{K}{2}Q^2, \quad K \equiv b + k.$$

Direct differentiation gives

$$(29) \quad \frac{\partial^2 W(Q; \omega)}{\partial Q^2} = -K < 0,$$

since $b > 0$ and $k > 0$. The planner's mean-CVaR functional is

$$(30) \quad \mathcal{W}(Q) = (1 - \lambda) \mathbb{E}[W(Q; \omega)] + \lambda \text{CVaR}_\alpha[W(Q; \omega)].$$

The expectation $\mathbb{E}[W(Q; \omega)] = \pi_H W(Q; H) + \pi_L W(Q; L)$ is a convex combination of the per-state welfares, each strictly concave by (29), hence strictly concave in Q . Under the discrete two-state representation, $\text{CVaR}_\alpha[W(Q; \omega)]$ is itself a convex combination of the per-state welfares (with non-negative weights summing to one — explicit formulae appear in (32) and (33) below), hence also strictly concave in Q . A convex combination of strictly concave functions is strictly concave, so $\mathcal{W}(Q)$ is strictly concave on $\mathbb{R}_+$ with second derivative $-K < 0$.

Step 2: Interior optimum. Writing $\mathcal{W}(Q) = A_{\text{eff}}Q - \frac{K}{2}Q^2$ where A_{eff} is the effective demand intercept derived in Step 3 of the proof of Theorem 1 (equations (34) and (35) below), the first-order condition gives

$$(31) \quad Q_{\text{risk}}^* = \frac{A_{\text{eff}}}{K}.$$

We show $A_{\text{eff}} > 0$ for all $\lambda \in [0, 1]$ and $\alpha \in (0, 1)$.

At $\lambda = 0$ both regime expressions reduce to $A_{\text{eff}} = \mathbb{E}[A] = \pi_H A_H + \pi_L A_L > A_L > 0$ by Assumptions 1, 4 and 5.

At $\lambda = 1$ in the collapsed regime ($\alpha \geq \pi_H$), (34) gives $A_{\text{eff}}|_{\lambda=1} = A_L > 0$. In the split regime ($\alpha < \pi_H$), (35) gives $A_{\text{eff}}|_{\lambda=1} = \mathbb{E}[A] - \frac{\alpha\pi_L}{1-\alpha}(w_H - w_L)$; expanding $\mathbb{E}[A] = \pi_H A_H + \pi_L A_L$ and using $\pi_H(w_H - w_L) = \mathbb{E}[A] - A_L$ yields $A_{\text{eff}}|_{\lambda=1} = A_L + \frac{(\pi_H - \alpha)}{1-\alpha}(w_H - w_L) > A_L > 0$ since $\alpha < \pi_H$.

For intermediate $\lambda \in (0, 1)$, A_{eff} is linear in λ with $\partial A_{\text{eff}}/\partial \lambda = -\pi_H(w_H - w_L) < 0$ in the collapsed regime and $-\frac{\alpha\pi_L}{1-\alpha}(w_H - w_L) < 0$ in the split regime; combined with the strictly positive endpoint values above, $A_{\text{eff}}(\lambda) \geq \min\{A_{\text{eff}}|_{\lambda=0}, A_{\text{eff}}|_{\lambda=1}\} > 0$.

Hence $Q_{\text{risk}}^* > 0$ is interior, and the non-negativity constraint $Q \geq 0$ is slack at the optimum — no Lagrangian–Karush–Kuhn–Tucker (KKT) bookkeeping is required.

Step 3: Sufficiency of the first-order condition (FOC). Strict concavity of $\mathcal{W}$ (Step 1) combined with the existence of an interior critical point (Step 2) implies the FOC is both necessary and sufficient for a global maximum, and that maximum is unique. $\square$

A.2. Theorem 1 (piecewise σ^* closed form)

PROOF OF THEOREM 1. *Step 1: Per-state welfare.* By the area-under-demand-minus-cost integral with linear inverse demand $p(Q; \omega) = w(\omega) - bQ$ (Assumption 2) and quadratic investment cost $C(Q) = \frac{k}{2}Q^2$ (Assumption 3),

the per-state social welfare is given by (28), reproduced here for reference:

$$W(Q; \omega) = (w(\omega) - c)Q - \frac{K}{2}Q^2.$$

Step 2: CVaR representation — two regimes. By the Rockafellar–Uryasev auxiliary-function representation (Rockafellar and Uryasev 2000), written in welfare form (equivalent to the loss-CVaR form $\inf_{\zeta} \{\zeta + (1 - \alpha)^{-1} \mathbb{E}[(L - \zeta)_+]\}$ under the substitution $L = -W$),

$$\text{CVaR}_\alpha[W(Q; \omega)] = \sup_{\zeta \in \mathbb{R}} \left\{ \zeta - \frac{1}{1-\alpha} \mathbb{E}[(\zeta - W(Q; \omega))_+] \right\},$$

which is the conditional mean of W in its worst- $1 - \alpha$ tail. The denominator $1 - \alpha$ is finite by Assumption 6. By Assumption 4 ($w_H > w_L$), the bad-tail event is unambiguously state L , since $W(Q; L) < W(Q; H)$ at any $Q > 0$. Two regimes arise from where the tail mass $1 - \alpha$ falls.

Regime A (collapsed): $\alpha \geq \pi_H$. The tail probability $1 - \alpha \leq \pi_L$ fits entirely inside state L , so the discrete CVaR collapses to the state- L welfare:

$$(32) \quad \text{CVaR}_\alpha[W(Q; \omega)] = W(Q; L) \quad (\alpha \geq \pi_H).$$

Regime B (split): $\alpha < \pi_H$. The tail mass exhausts all of state L (probability π_L) plus a slice of size $1 - \alpha - \pi_L = \pi_H - \alpha$ of state H . The probability-weighted tail average is

$$(33) \quad \text{CVaR}_\alpha[W(Q; \omega)] = \frac{\pi_L W(Q; L) + (\pi_H - \alpha) W(Q; H)}{1 - \alpha} \quad (\alpha < \pi_H).$$

Step 3: Effective demand intercept. Substitute (32) or (33) into the mean-CVaR objective (30) and collect the Q -coefficient.

Regime A. The effective probability on state L is $\pi_L + \lambda \pi_H$ and on state H is $(1 - \lambda)\pi_H$. Hence

$$(34) \quad \begin{aligned} A_{\text{eff}}^{\text{coll}} &= (\pi_L + \lambda \pi_H)(w_L - c) + (1 - \lambda)\pi_H(w_H - c) \\ &= \pi_H(w_H - c) + \pi_L(w_L - c) - \lambda \pi_H[(w_H - c) - (w_L - c)] \\ &= \mathbb{E}[A] - \lambda \pi_H (w_H - w_L). \end{aligned}$$

Regime B. The effective probability on state H becomes $(1 - \lambda)\pi_H + \lambda(\pi_H - \alpha)/(1 - \alpha)$, which simplifies to $\pi_H - \frac{\lambda \alpha \pi_L}{1 - \alpha}$ using $\pi_L = 1 - \pi_H$. The effective probability on state L is the complement, $\pi_L + \frac{\lambda \alpha \pi_L}{1 - \alpha}$. Collecting the Q -coefficient gives

$$(35) \quad A_{\text{eff}}^{\text{split}} = \mathbb{E}[A] - \lambda \frac{\alpha \pi_L}{1 - \alpha} (w_H - w_L).$$

Step 4: FOC and risk-averse optimum. By Lemma 1, the planner's objective $\mathcal{W}(Q) = A_{\text{eff}} Q - \frac{K}{2}Q^2$ is strictly concave in Q with interior maximum. The first-order condition $\partial \mathcal{W} / \partial Q = A_{\text{eff}} - KQ = 0$ delivers

$$(36) \quad Q_{\text{risk}}^* = \frac{A_{\text{eff}}}{K},$$

with A_{eff} given by (34) or (35) according to regime. The risk-neutral benchmark $\lambda = 0$ reduces both expressions to $A_{\text{eff}} = \mathbb{E}[A]$, so

$$(37) \quad Q_{\text{RN}}^* = \frac{\mathbb{E}[A]}{K}.$$

Step 5: Subsidy that restores the risk-neutral optimum. A per-unit production subsidy σ shifts the committing agent's effective intercept by σ : by Assumption 9, the credit enters the agent's marginal return identically to a price increment while its funding stays outside the mean-CVaR aggregate (were it financed inside the aggregate, the transfer would net out state by state and leave the optimum unchanged). The subsidy that restores Q_{RN}^* solves $A_{\text{eff}} + \sigma^* = \mathbb{E}[A]$. Substituting (34) and (35) yields

$$(38) \quad \sigma_{\text{coll}}^* = \lambda \pi_H (w_H - w_L) \quad (\alpha \geq \pi_H),$$

$$(39) \quad \sigma_{\text{split}}^*(\alpha) = \lambda \frac{\alpha \pi_L}{1 - \alpha} (w_H - w_L) \quad (\alpha < \pi_H).$$

Both are non-negative under Assumptions 4 and 7. Continuity at the boundary $\alpha = \pi_H$ is established in Corollary 3 below. Combining the two regimes into a single closed form,

$$(40) \quad \sigma^*(\alpha) = \lambda (w_H - w_L) \cdot \min\left\{\frac{\alpha \pi_L}{1 - \alpha}, \pi_H\right\}.$$

The risk-neutral limit $\lambda = 0$ gives $\sigma^* = 0$ for all α ; the fully tail-focused limit $\lambda = 1$ gives the upper bound $\sigma^*(\alpha) = \pi_H (w_H - w_L)$ in the collapsed regime. $\square$

A.3. Corollary 3 (continuity at $\alpha = \pi_H$)

PROOF OF COROLLARY 3. Evaluate the split-regime expression (39) at the boundary $\alpha = \pi_H$, using $1 - \pi_H = \pi_L$ from Assumption 1:

$$\sigma_{\text{split}}^*(\pi_H) = \lambda \frac{\pi_H \pi_L}{1 - \pi_H} (w_H - w_L) = \lambda \frac{\pi_H \pi_L}{\pi_L} (w_H - w_L) = \lambda \pi_H (w_H - w_L) = \sigma_{\text{coll}}^*.$$

Hence the left and right limits of $\sigma^*(\alpha)$ at $\alpha = \pi_H$ coincide, establishing continuity on $(0, 1)$.

The one-sided derivatives differ. Differentiating (39) with respect to α ,

$$\frac{d\sigma_{\text{split}}^*}{d\alpha} = \frac{\lambda \pi_L (w_H - w_L)}{(1 - \alpha)^2},$$

so the left derivative at the boundary is

$$(41) \quad \sigma^{*'}(\pi_H^-) = \frac{\lambda \pi_L (w_H - w_L)}{(1 - \pi_H)^2} = \frac{\lambda (w_H - w_L)}{\pi_L} > 0,$$

while (38) is constant in α , giving right derivative $\sigma^{*'}(\pi_H^+) = 0$. The mismatch (41) vs. 0 is a benign kink in $\sigma^*(\alpha)$, not a discontinuity. $\square$

A.4. Corollary 2 ($\text{EVPI} = \text{Var}(A)/(2K)$)

PROOF OF COROLLARY 2. We compute the expected value of perfect information for the risk-neutral planner ($\lambda = 0$), the textbook EVPI baseline used in Section 2.3 of the main text.

Step 1: Wait-and-see optimum per scenario. For each realized state ω , the per-state planner's problem $\max_Q W(Q; \omega)$ has FOC $(w(\omega) - c) - KQ = 0$, so the wait-and-see output is

$$(42) \quad Q^*(\omega) = \frac{w(\omega) - c}{K} = \frac{A}{K}.$$

By the envelope theorem (Mas-Colell, Whinston, and Green 1995, Theorem M.L.1), substituting (42) into (28) delivers the optimal value

$$(43) \quad W^*(\omega) = \frac{(w(\omega) - c)^2}{2K} = \frac{A^2}{2K}.$$

Step 2: Recourse optimum at the mean. The risk-neutral recourse plan uses $Q_{\text{RN}}^* = \mathbb{E}[A]/K$ from (37). Substituting into $\mathbb{E}_\omega[W(Q_{\text{RN}}^*; \omega)] = \mathbb{E}[A] \cdot Q_{\text{RN}}^* - \frac{K}{2}(Q_{\text{RN}}^*)^2$ yields

$$(44) \quad W^*(\mathbb{E}[A]) = \frac{(\mathbb{E}[A])^2}{2K}.$$

(Use linearity of expectation: $\mathbb{E}[A \cdot Q_{\text{RN}}^*] = Q_{\text{RN}}^* \mathbb{E}[A] = (\mathbb{E}[A])^2/K$.)

Step 3: EVPI = Var(A)/(2K). Following the Madansky (1960) inequality framework, the expected value of perfect information is

$$\text{EVPI} \equiv \mathbb{E}_\omega[W^*(\omega)] - W^*(\mathbb{E}[A]).$$

Substituting (43) and (44),

$$(45) \quad \begin{aligned} \text{EVPI} &= \frac{\mathbb{E}[A^2]}{2K} - \frac{(\mathbb{E}[A])^2}{2K} \\ &= \frac{\mathbb{E}[A^2] - (\mathbb{E}[A])^2}{2K} \\ &= \frac{\text{Var}(A)}{2K}. \end{aligned}$$

The display in (45) matches the textbook quadratic-objective expression in Birge and Louveaux (2011, §4.7).

Specialized to the two-state distribution under Assumption 1, $\text{Var}(A) = \pi_H \pi_L (w_H - w_L)^2$, so

$$(46) \quad \text{EVPI} = \frac{\pi_H \pi_L (w_H - w_L)^2}{2K}.$$

□

A.5. Proposition 1 (additive credit decomposition)

Throughout this subsection and the next, write $m_\alpha \equiv \min\{\alpha\pi_L/(1-\alpha), \pi_H\}$, so that the two regime expressions (34) and (35) combine into the single form

$$(47) \quad A_{\text{eff}} = \mathbb{E}[A] - \lambda m_\alpha (w_H - w_L),$$

and $\sigma^*(\alpha) = \lambda m_\alpha (w_H - w_L)$ by (40).

PROOF OF PROPOSITION 1. *Step 1: First-best output.* With the externality internalized, per-state social welfare is $W_v(Q; \omega) = (w(\omega) + v - c)Q - \frac{K}{2}Q^2$. The risk-neutral first-best therefore replaces $\mathbb{E}[A]$ with $\mathbb{E}[A] + v$ in (37): $Q_{\text{FB}}^* = (\mathbb{E}[A] + v)/K$, establishing (i).

Step 2: The committing agent's uncorrected choice. The mean-CVaR agent who does not internalize v chooses $Q = A_{\text{eff}}/K$ with A_{eff} as in (47): the state-independent constant v appears in neither the market expectation nor the market tail mean she prices.

Step 3: Restoring credit. A per-unit credit σ_{tot} enters the committing agent's marginal return identically to a price increment (Step 5 of the proof of Theorem 1), so the credit that delivers Q_{FB}^* solves $A_{\text{eff}} + \sigma_{\text{tot}} = \mathbb{E}[A] + v$. Substituting (47),

$$\sigma_{\text{tot}} = v + \lambda m_\alpha (w_H - w_L) = v + \sigma^*(\alpha),$$

establishing (ii). For (iii), $\partial\sigma^*(\alpha)/\partial v = 0$ because v cancels from $\mathbb{E}[A] - A_{\text{eff}}$, and $\partial v/\partial(\lambda, \alpha, \pi_H, w_H - w_L) = 0$ because v is a primitive of the externality, not of the price distribution. No cross term appears because (47) is affine in v with unit coefficient. $\square$

A.6. Proposition 2 and Corollary 4 (instrument form)

PROOF OF PROPOSITION 2. *Part (i): flat credit.* Immediate from Step 5 of the proof of Theorem 1: the credit $\sigma^*(\alpha)$ restores Q_{RN}^* , and it is paid on every unit in every state, so the expected per-unit outlay is $T_{\text{flat}} = \sigma^*(\alpha)$.

Part (ii): one-sided floor. Fix a strike $\kappa \in [w_L, w_H)$. The floor pays $\kappa - w(\omega)$ per unit exactly when $w(\omega) < \kappa$, i.e. in state L only, so the post-floor economy is the two-state economy with intercepts (w_H, κ) , unchanged probabilities, and unchanged state ordering ($\kappa < w_H$). The regime classification in Step 2 of the proof of Theorem 1 depends only on (α, π_H) , so (47) applies with the floored primitives:

$$A_{\text{eff}}^{\text{fl}}(\kappa) = \mathbb{E}[A] + \pi_L(\kappa - w_L) - \lambda m_\alpha (w_H - \kappa),$$

using $\mathbb{E}^{\text{fl}}[A] = \pi_H(w_H - c) + \pi_L(\kappa - c) = \mathbb{E}[A] + \pi_L(\kappa - w_L)$. The restoring condition $A_{\text{eff}}^{\text{fl}}(\kappa^*) = \mathbb{E}[A]$ reads

$$(48) \quad \pi_L(\kappa^* - w_L) = \lambda m_\alpha (w_H - \kappa^*),$$

whose unique solution is the convex combination

$$\kappa^* = \frac{\pi_L w_L + \lambda m_\alpha w_H}{\pi_L + \lambda m_\alpha},$$

which lies in (w_L, w_H) whenever $\lambda m_\alpha > 0$ (and equals w_L , a vacuous floor, at $\lambda = 0$, consistent with $\sigma^* = 0$).

The expected per-unit outlay is the state- L payment weighted by its probability:

$$T_{\text{floor}} = \pi_L (\kappa^* - w_L) = \frac{\pi_L \lambda m_\alpha (w_H - w_L)}{\pi_L + \lambda m_\alpha} = \sigma^*(\alpha) \cdot \frac{\pi_L}{\pi_L + \lambda m_\alpha},$$

where the middle equality substitutes $\kappa^* - w_L = \lambda m_\alpha (w_H - w_L) / (\pi_L + \lambda m_\alpha)$ from (48).

Part (iii): two-sided CfD. Under the swap at strike $\bar{w} = \mathbb{E}[w(\omega)]$, the per-state, per-unit margin becomes $w(\omega) + (\bar{w} - w(\omega)) - c = \bar{w} - c$, deterministic. For a degenerate distribution, $\text{CVaR}_\alpha[W] = \mathbb{E}[W] = W$ at every $\alpha \in (0, 1)$, so the mean-CVaR objective collapses to $(\bar{w} - c)Q - \frac{K}{2}Q^2$ for every $\lambda \in [0, 1]$, with maximizer $Q = (\bar{w} - c)/K = \mathbb{E}[A]/K = Q_{\text{RN}}^*$. The expected per-unit outlay is $T_{\text{CFD}} = \mathbb{E}[\bar{w} - w(\omega)] = 0$ by construction.

Ranking. For $\lambda \in (0, 1]$: $m_\alpha > 0$ under Assumptions 1 and 6, so $T_{\text{floor}} > 0 = T_{\text{CFD}}$ and $T_{\text{floor}}/T_{\text{flat}} = \pi_L/(\pi_L + \lambda m_\alpha) \in (0, 1)$, giving $0 < T_{\text{floor}} < T_{\text{flat}}$. $\square$

PROOF OF COROLLARY 4. Value net public transfers state-by-state at a *state-independent* shadow price of public funds $\mu \geq 1$ (a constant marginal cost of public funds is maintained throughout; a state-dependent $\mu(\omega)$ would add a covariance term to the two-sided CfD's deadweight cost and is outside the corollary's scope): a transfer of $t(\omega)$ per unit to the chain in state ω costs society $(\mu - 1)t(\omega)$ per unit beyond the transfer itself (a negative $t(\omega)$ — the high-state leg of the two-sided CfD — is a receipt and carries the symmetric shadow saving). All three instruments in Proposition 2 induce the same output Q_{RN}^* and the same market surplus, so welfare under instrument i differs from welfare under instrument j only through the expected deadweight term:

$$\Delta W_{ij} = (\mu - 1)(T_j - T_i)Q_{\text{RN}}^*,$$

by linearity of the expectation in $t(\omega)$. At $\mu = 1$ the term vanishes and the instruments are welfare-equivalent; at $\mu > 1$ the ranking of Proposition 2 becomes strict in welfare. $\square$

B. Aggregation Invariance

PROPOSITION 3 (Aggregation invariance of the harvest-cost term). *Let the hex-level Stage-1 objective use harvest-cost coefficient $c_h = \kappa \cdot \text{HC}/(\text{AvailInv}_h + \varepsilon)$, where $\kappa, \text{HC} > 0$, $\varepsilon \geq 0$, and AvailInv_h is hex h 's available Year-0 residue inventory. Define county aggregates $\text{AvailInv}_F = \sum_h \theta_{h,F} \cdot \text{AvailInv}_h$ and $K_F = \sum_h \theta_{h,F}$, where $\theta_{h,F}$ is the area share of hex h in county F with $\sum_F \theta_{h,F} = 1$, and set $c_F = \kappa \cdot \text{HC} \cdot K_F/(\text{AvailInv}_F + \varepsilon)$. Assume (A1) harvest allocations are proportional to availability with a common factor, $\bar{x}_h = \gamma \text{AvailInv}_h$ and $\bar{x}_F = \gamma \text{AvailInv}_F$ for some $\gamma > 0$ (the unconstrained Stage-1 first-order condition); and (A2) $\varepsilon = 0$, or $\text{AvailInv} \gg \varepsilon$ at every node. Then the harvest-cost totals coincide: $\sum_F c_F \cdot \bar{x}_F = \sum_h c_h \cdot \bar{x}_h$ (exactly under $\varepsilon = 0$; up to $O(\varepsilon)$ otherwise).*

PROOF. Under (A1) and $\varepsilon = 0$, $c_h \bar{x}_h = \kappa \text{HC} \gamma$ for every hex, so $\sum_h c_h \bar{x}_h = \kappa \text{HC} \gamma |H|$. At the county level, $c_F \bar{x}_F = \kappa \text{HC} \gamma K_F$, and $\sum_F K_F = \sum_F \sum_h \theta_{h,F} = \sum_h \sum_F \theta_{h,F} = |H|$, using $\sum_F \theta_{h,F} = 1$. Hence $\sum_F c_F \bar{x}_F = \kappa \text{HC} \gamma |H| = \sum_h c_h \bar{x}_h$. For $\varepsilon > 0$ the two sides differ by a Jensen-type term in the within-county dispersion of AvailInv , which is $O(\varepsilon)$ under (A2) and vanishes as $\varepsilon \rightarrow 0$. $\square$

Scope. The proposition covers the harvest-cost term only. Hypothesis (A1) is the unconstrained Stage-1 first-order condition; it fails where depot minimum-throughput, capacity, or ash-share constraints bind, as they do in the

Stage-2 MILP. Transport aggregation uses the distance-averaging rules below, a linearization rather than an identity; revenue, OPEX, CAPEX, and the consumer-surplus integral aggregate by summation. The operative bound for the full accounting stack is therefore the empirical residual of $\pm 0.15\%$ reported in Table 4, not the exact identity.

Construction. The Stage-2 operational solve runs at the FIPS-5 county level (five-digit Federal Information Processing Standards codes) rather than the underlying 50 mi^2 hex resolution. Because 8,864 of 14,040 hexes straddle multiple counties, hex-level quantities are split across counties proportional to area shares $\alpha_{h,F} \equiv \text{area}(h \cap F) / \text{area}(h)$, with $\sum_F \alpha_{h,F} = 1$ for every hex h . The aggregation rules for the five Stage-1 fields are:

$$\begin{aligned} \text{inv}_F[w] &= \sum_h \alpha_{h,F} \cdot \text{inv}_h[w] && \text{(area-attribute)} \\ \text{growth}_F[w] &= \sum_h \alpha_{h,F} \cdot \text{growth}_h[w] && \text{(area-attribute)} \\ \text{ratio}_F &= \frac{\sum_h \alpha_{h,F} \cdot \text{ratio}_h \cdot \text{inv}_h^{HW}}{\sum_h \alpha_{h,F} \cdot \text{inv}_h^{HW}} && \text{(HW-inventory-weighted)} \\ \text{dem}_F[w] &= \sum_{m: \text{maj}(m)=F} \text{dem}_m[w] && \text{(point-facility majority)} \\ d_{F,F'}^{HM} &= \frac{\sum_{(h,m): \text{maj}(m)=F'} d_{h,m} \cdot \alpha_{h,F}}{\sum_{(h,m): \text{maj}(m)=F'} \alpha_{h,F}} && \text{(hex-share + mill-majority)} \end{aligned}$$

The key step is the K-factor cost normalization: the hex-level Stage-1 LP objective coefficient is $c_{h,w} = HC_w \cdot 6.25 \cdot 3200 / (\text{AvailInv}_{h,w}^{Y_0} + \epsilon)$; naïvely aggregating the denominator across hexes shrinks the objective by $\sim K = \text{avg hexes/county}$. We restore invariance by setting

$$c_{F,w} = HC_w \cdot \frac{6.25 \cdot 3200 \cdot K_F}{\text{AvailInv}_{F,w}^{Y_0} + \epsilon}, \quad K_F \equiv \sum_{h \in P} \alpha_{h,F}.$$

Under the LP-optimal first-order condition $x_F[w] \propto \text{AvailInv}_{F,w}$, this gives $\sum_F c_{F,w} x_F[w] = \sum_h c_{h,w} x_h[w]$, so county-level and hex-level total costs coincide.

Feedstock envelope. Two envelope parameters referenced in Section 4 are fixed as follows. Transport catchments are capped at $H2D_{\max} = D2R_{\max} = 100 \text{ mi}$ for the source-to-depot and depot-to-refinery legs respectively. The source-to-depot leg is widened from the 50-mi convention of the antecedent siting literature because the sustainable-removal cut below shrinks per-county recoverable residue, and the 200 kdt/yr depot minimum-throughput constraint requires a wider draw to remain feasible at Grand Challenge scale; the depot-to-refinery leg is widened from 75 mi to 100 mi for symmetry with the first leg. Recoverable residue is scaled by a sustainable-removal fraction of 0.70, the U.S. Department of Energy (DOE) biomass scoping convention (U.S. Department of Energy 2016), applied on top of the willingness-to-harvest ($W2HF = 0.28$) and commercial-private share ($\text{ComR} = 0.338$) filters of Section 4.

Table 4: Hex-level vs. county-level totals at the deterministic baseline

Metric	Hex-level	County-level	Relative diff.
Total NPV cost (10-yr)	\$98.21 B	\$98.36 B	+0.15%
Total AResidue (pre-SRU, 10-yr cum.) ^a	1.205 B dt	1.204 B dt	−0.07%
Total stumpage cost (TSC)	\$29.95 B	\$29.95 B	0.00%

Notes: Aggregation invariance holds within 0.2% across all three accounting metrics.

^a The AResidue row is a hex-vs-county aggregation-invariance check computed *before* the 0.70 sustainable-removal (SRU) scaling, as a 10-year cumulative of a growing series; it is not a supply statement. On a like-for-like post-SRU basis the comparison is 0.844 vs. 0.843 B dt (−0.07%; the common factor cancels from the relative difference). The operational Stage-2 model consumes Year-0 availability only: 72.5 M dt (Section 4).

Numerical sanity check. At the deterministic baseline ($p_{\text{SAF}} = \$5.97/\text{gal}$, 10-year horizon), the hex vs. county discrepancies are:

Per-state breakdowns can shift by up to $\pm 10\%$ (the LP re-allocates harvest within state to exploit county-level cost variation), but the total and the state-ranked Top-3 are preserved.

Table 5 records the dimensions of the aggregated instance that the rest of the paper solves.

C. KDE Discretization

We fit a Gaussian kernel density estimator (KDE) on the constructed drop-in biofuel price series described in Section 4: 3,046 daily-interpolated observations from weekly sources, August 2013 through December 2021.

Construction and provenance. The series is built in three steps, inherited from the producer-revenue calibration of Li et al. (2026). (i) Weekly U.S. retail gasoline and diesel prices (U.S. Energy Information Administration 2024) are slate-weighted and adjusted to a common real basis by a compounded real-interest parameter, giving the fuel-market component (sample mean \$3.10/gal). (ii) Weekly D-code RIN prices are scaled by the 1.7 cellulosic equivalence factor and the same real adjustment, giving the RFS compliance-value component (sample mean \$2.96/gal); renewable jet equivalence values are in practice assigned facility-by-facility at 1.6–1.7 with most at 1.6 (Section E), and re-scaling to 1.6 would move the mean to $\approx \$5.88/\text{gal}$, within the robustness of every downstream object. (iii) The revenue price is their sum, mean \$5.98/gal with standard deviation \$1.27/gal. The raw workbook, construction script, and merged series are part of the replication package with source identifiers and download dates. Two properties of the construction are load-bearing and are stated here rather than left for the reader to discover. First, the price is *policy-inclusive*: roughly half of its level is RFS compliance value, so the non-capitalization hypothesis of Proposition 1 fails with respect to RFS support, and Section 5.5 re-states the implied-SCC inversion accordingly (the baseline reading is *incremental to the existing RIN stack*). Second, the dispersion that the risk objects price is genuine producer revenue risk: a producer paid partly in credits bears RIN volatility one-for-one, and the two-state spread $w_H - w_L = \$2.19/\text{gal}$ reflects joint fuel-price and credit-price variation over the sample. The weekly source data are serially persistent and the daily interpolation adds no information; refitting the KDE on the 436 weekly source observations moves the collapse primitives little ($\pi_H = 0.44$, spread \$2.20, $\sigma^* = \$0.48$ against the published \$0.52), and halving or doubling the Silverman bandwidth moves σ^* only within \$0.48–\$0.53. Persistence matters more than frequency. The scenario structure

Table 5: Spatial and temporal scope of the SE-U.S. residue-to-SAF supply-chain MILP

Quantity	Value
<i>Spatial frame</i>	
Hexagons (50 mi ² , SE-US frame)	14,040
Counties (FIPS-5, after hex → county agg.)	1,374
Mills $ \mathcal{M} $ (Stage-1 wood demand)	1,663
Active residue-source counties $ \mathcal{H} $ (Year-0, hardwood)	220
Depot candidate counties $ \mathcal{D} $ (528 industrial hexes)	483
Refinery candidate counties $ \mathcal{R} $	483
<i>Eligible edge sets (operational cutoffs, county level)</i>	
Stage-1 hex → mill pairs (≤ 100 mi)	244,634
Stage-2 residue → depot pairs (≤ 100 mi)	6,600
Stage-2 depot → refinery pairs (≤ 100 mi)	11,314
<i>Time and scenarios</i>	
Planning horizon (Year ₀ . . . Year ₉)	10 yrs
SAF-price scenarios $ \mathcal{S} $ (KDE bins)	20
Ash-tier categories ($A_d \times A_r$)	10×10

Notes: Hex resolution is 50 mi² (U.S. Forest Service, USFS, Wood Resources frame); the operational Stage-2 solve aggregates to the FIPS-5 county level (five-digit Federal Information Processing Standards codes). Both Stage-2 catchment legs are capped at 100 mi (widened from the 50/75-mi hex-model convention so that depot minimum-throughput constraints remain feasible at county resolution; Section B). Pair counts are county-level edge counts after the cutoff is imposed. Variable and constraint counts, the mixed-integer programming (MIP) gap target, and realized gaps are reported separately in Table 9.

of the two-stage program treats the realized price as a persistent decade-scale state, the reading under which the spot dispersion (spread \$2.19) is the relevant exposure. Under the opposite extreme — rapid mixing, so the committed chain earns the multi-year *average* price — a moving-block bootstrap of H -year averages (26-week blocks, 2,000 draws) gives spreads of \$1.34, \$0.58, and \$0.41 at $H = 1, 5, 10$ and σ^* of \$0.35, \$0.14, and \$0.11. The risk component therefore carries the persistence assumption openly: \$0.52 under permanent price states, falling toward \$0.11 under full decade-scale mean reversion. An 8.4-year sample cannot pin the long-run persistence of the fuel-plus-RIN price level, and we flag this as the primary calibration uncertainty on the risk side (Section 6.3).

Identification: mapping the constructed price to SAF revenue. SAF is not yet a deep liquid market: in 2026, the modal SAF-buyer is an airline operating under bespoke ESG offtake contracts (United-NEXT, Delta–Air-Products, Microsoft–Twelve) at premia of approximately \$2–\$4/gal above wholesale jet-A. The constructed series proxies the revenue of a drop-in fuel producer paid the fuel-market price plus compliance-credit value; for SAF the corresponding stack is jet-A plus the credit-and-premium package, of comparable magnitude to the constructed mean. The slack-import term $p^{\text{jet}} \cdot s(\omega)$ in (3) remains a fossil-jet benchmark on a policy-free basis (Section 2.2), so the make-or-buy comparison prices domestic production inclusive of its policy revenue against fossil import at its market price; the welfare interpretation of the RIN component — transfer versus resource value — is taken

up where the decomposition uses the price level (Section 5.5). Two first-order robustness notes: σ^* depends on the state spread $w_H - w_L$, which is robust to additive shifts of the level; the EVPI level scales linearly with $\text{Var}(p_{\text{SAF}})$ and would shift with the (unobserved) variance of a mature SAF market relative to the constructed series.

Evaluation vintages, certification, and convergence. The per-scenario statistics of Section 5.2 come from a certified evaluation computed under a single data state. Two vintages preceded it, and each correction moved the estimate the same way. (i) An early vintage — retained in the replication package — carried low-price wait-and-see gaps of 27–79% and required a dominance-certificate correction (raw $\mathbb{E}[L] = \$97.7$ M/yr versus conservative \$169 M/yr). (ii) A refreshed vintage re-solved the six lowest-price bins under a 7,200 s limit with small-topology warm starts (realized gaps 1.3–5.0%) and gave $\mathbb{E}[L] = \$185$ M/yr; but its own candidate evaluations exposed the incumbency trap one level up: the 197/27 plan’s fixed-siting values exceeded the recorded wait-and-see incumbents in bins 2 and 4–9, by \$48 M to \$432 M — negative raw regrets that are certificates of wait-and-see suboptimality, not noise to be floored away. (iii) The certified vintage re-solves those seven bins warm-started from the certifying topology; the recorded wait-and-see values now dominate every fixed-siting evaluation in the candidate pool, realized gaps are 1.3–3.9%, and $\mathbb{E}[L] = \$299$ M/yr; every here-and-now value quoted in the text is a fixed-siting solve with recorded gap below 10^{-12} . Each step raises the floor of a lower-bound band whose gap-implied ceiling is \$827 M/yr (Section 5.2). Known artifacts of the evaluation files — including one superseded candidate whose rows duplicate the central plan and are excluded from every statistic — are documented in the replication package.

Calibration choice for the regret evaluation. An earlier evaluation paired the elastic-side anchor $\varepsilon = -1.0$ with a \$5.68/gal voluntary-premium anchor. At that calibration the low-price wait-and-see MILPs do not improve on the warm start within the time budget — the bin-1 solve terminates at a 79% residual gap — and the recorded regret lower bound degenerates to \$4.5 M/yr, a convergence artifact rather than an economic finding. The $\varepsilon = -0.5$ evaluation reported in Section 5.2 is the one whose low-price solves move far enough off the warm start to be informative, including the certified low-price re-solves of vintage (iii) above.

The KDE bandwidth follows Silverman’s rule of thumb (Silverman 1986),

$$h^* = 1.06 \cdot \hat{\sigma} \cdot n^{-1/5},$$

robust to mild non-Gaussianity in this constructed price series. Sheather–Jones (Sheather and Jones 1991) and Wand–Jones (Wand and Jones 1995) plug-in alternatives shift the bandwidth by $\pm 7\%$; the 20-bin discretization absorbs this variation (σ_{KDE}^* moves by $< 3\%$ across the three bandwidth rules).

We discretize the continuous KDE into 20 equal-width bins over the empirical support [\$3.81, \$9.12]/gal, assigning each bin its KDE-weighted probability p^ω and conditional mean $p_{\text{SAF}}(\omega) = \mathbb{E}[p_{\text{SAF}} \mid p_{\text{SAF}} \in \text{bin}_\omega]$ (Table 8). The raw weekly sample has mean \$5.97/gal and standard deviation \$1.27/gal; the 20-bin discretization has probability-weighted mean $\mathbb{E}[p_{\text{SAF}}] = \$6.00/\text{gal}$ and standard deviation \$1.28/gal, and all discretization-derived quantities in the paper use the binned values. The bin count of 20 matches the typical precision-vs.-tractability trade-off in scenario-based SP (Birge and Louveaux 2011) and exactly resolves the empirically relevant tail at $\alpha = 0.95$ (cf. Table 8: $p^{19} + p^{20} = 0.012$, which sits cleanly within the $1 - \alpha = 0.05$ CVaR confidence window).

Table 6: The candidate pool of the risk evaluation: provenance and per-plan statistics at the refined 81-breakpoint grid

Plan	Bin	Vint.	N_D/N_R	Prov. gap	$\mathbb{E}[L]$	CVaR ₉₅ [L] (\$M/yr)	$\mathbb{E}[W]$	CVaR ₉₅ [W] (\$B/yr)
committed ^a	9	(i)	248/34	18.12% ^b	300	931	17.004	11.426
tight01	1	(ii)	192/26	2.04%	626	1,804	16.678	12.351
tight02	2	(ii)	195/26	3.86%	949	2,337	16.355	12.134
tight03	3	(ii)	197/27	1.34%	400	1,837	16.904	12.353
tight04	4	(ii)	204/28	2.14%	369	2,062	16.935	12.197
tight05	5	(ii)	211/30	3.91%	508	1,334	16.797	11.959

Notes: Statistics from LP-exact fixed-siting recourse evaluations of each plan across all 20 KDE bins at an 81-breakpoint SOS2 grid, against the pool-maximum wait-and-see column (Section 3.5); the 21-breakpoint evaluation’s cross-plan margins are chord-error artifacts (Section 5.4). Vintage labels follow the evaluation-history paragraph above. The committed plan minimizes $\mathbb{E}[L]$ and CVaR₉₅[L] and maximizes $\mathbb{E}[W]$; tight03 maximizes CVaR₉₅[W]; plans 195/26, 204/28, and 211/30 are no longer scenario-optimal after certification and are retained as candidates only. The vintage-(ii) bin-6 solve is bit-identical to the committed plan and is not listed separately.

^a Provenance: deterministic bin-9 sweep; the identical siting is reproduced by the bin-6 endogenous-demand solve at 4.81%.

^b On the fixed-demand PVC objective of the provenance sweep.

Bin-count sensitivity. We test the robustness of the 20-bin choice by re-aggregating the KDE at $M \in \{10, 20, 40\}$ bins and recomputing the 2-state collapse of Section 2.4. At $M = 10$ (pair-merging consecutive bins), the KDE-collapsed $\sigma_{\text{KDE}}^* = \$0.518/\text{gal}$ and EVPI at $K = 1$ is $\$0.600/\text{gal}^2$, matching the $M = 20$ baseline values exactly because pair-merging preserves the median-split conditional means. At $M = 40$ (splitting each bin at its midpoint under uniform-within-bin densities), σ_{KDE}^* shifts to $\$0.490/\text{gal}$ and EVPI to $\$0.610/\text{gal}^2$ — a 5.4% drop in σ^* and a 1.7% rise in EVPI relative to baseline. The 20-bin choice is therefore a stable representative of the KDE for both headline quantities.

D. Computational Details

Source runs for the committed plan. The committed here-and-now reference of Section 5 is the bin-9 central-scenario solve (248 depots / 34 refineries, 3.32 B gal/yr, \$1.43 B/yr chain surplus); a companion re-solve at the raw mean price of \$5.97/gal places 228/31 (3.07 B gal/yr, \$1.20 B/yr). The \$0.08/gal difference between the two calibrations straddles the topology margin at the lower edge of the expansion band in Figure 8, which is why the two runs differ in siting. Both terminate at the wall-clock limit with realized relative gaps of 18.12% (bin 9) and 15.20% (\$5.97), so the reported topologies are feasible incumbents rather than certified optima; both runs are retained in the replication package.

Provenance of the candidate pool. The committed plan’s 18.12% gap is on the fixed-demand present-value objective. Its siting is reproduced exactly by the bin-6 endogenous-demand wait-and-see solve at a 4.8% gap, so on the objective actually used in Section 5.4 the committed topology is a converged candidate rather than a wide-gap incumbent. The pool contains no plan larger than 248/34: no wait-and-see re-solve, warm-started from either the committed or a contracted topology, returned a larger siting in any bin. The joint mean-CVaR MILP is not solved (Section 6.3).

Table 7: Per-bin certified wait-and-see values, realized gaps, and gap-implied regret allowance (endogenous-demand evaluation)

Bin	p^ω	W^{WS} (\$B)	gap	L_s (\$M)	allowance $p^\omega g_\omega W^{\text{WS}}$ (\$M)
1	0.0328	12147914935.756	2.0%	947	8.1
2	0.0593	12723909183.871	2.9%	872	21.5
3	0.0750	13308753337.244	1.3%	781	13.4
4	0.0749	13925039652.988	2.8%	671	29.2
5	0.0675	14579488394.909	2.9%	554	28.4
6	0.0671	15269195972.772	3.0%	432	31.0
7	0.0743	15877584179.751	3.2%	325	37.5
8	0.0772	16578743018.242	3.5%	180	44.7
9	0.0635	17118613056.083	3.9%	48	42.0
10	0.0444	17951363404.618	3.7%	0	29.5
11	0.0395	18755595944.940	3.3%	0	24.8
12	0.0481	19554799375.917	3.1%	0	29.5
13	0.0581	20253857812.329	3.1%	0	36.2
14	0.0607	21101838682.476	3.1%	0	39.1
15	0.0565	21868292880.931	3.1%	0	37.7
16	0.0454	22667461822.222	3.1%	0	31.8
17	0.0289	23429918450.411	3.1%	0	21.3
18	0.0148	24293572821.621	3.2%	0	11.7
19	0.0079	25023687891.312	3.5%	0	6.9
20	0.0042	25759765497.633	3.7%	0	4.0
Expectation					528

Notes: g_ω is the realized relative gap of the recorded wait-and-see value. The allowance column sums to the \$528M gap-implied expectation quoted in Section 5.2; because the certified wait-and-see column coincides with the candidate-pool maximum in every bin, the allowance in bins 10–20 is capacity for unrecorded regret outside the pool, not evidence of it.

Breakpoint-grid refinement. All cross-plan statistics in Section 5.4 are computed at an 81-breakpoint SOS2 grid (LP-exact per bin). The refinement was not cosmetic. At the original 21-breakpoint grid the 197/27 plan recorded a +\$15 M/yr expected-welfare advantage over the committed plan; at 41 and 81 breakpoints that margin reverses to −\$99.5 M/yr, with the two refined values agreeing to \$0.1 M — the $O(\Delta Q^2)$ convergence signature of the piecewise-linear chord error. The coarse grid’s cross-plan margins were therefore chord-error artifacts, exactly the failure mode the linearization invites on margins of this size. The committed plan’s own statistics barely move under the same refinement ($\mathbb{E}[L]$: \$299 to \$300 M/yr; $\text{CVaR}_{0.95}[L]$: \$921 to \$931 M/yr; tail-to-mean 3.1), so the regret statistics of Section 5.2 are grid-robust while the cross-plan comparison at the coarse grid was not. The Stage-2 problem is solved separately for each of the 20 KDE scenarios in the candidate-evaluation spirit of Birge and Louveaux (2011) (Ch. 4). The Mak, Morton, and Wood (1999) and Bayraksan and Morton (2011) replication-based schemes would deliver a *statistical* upper bound on $\mathbb{E}[W^*]$ via Monte Carlo replications of the scenario set; we do not run replications, and the deterministic 20-bin discretization does not by itself deliver a certified bound either, since the per-scenario values entering the expectation are incumbents rather than best bounds. What the reported \$255 M estimate is — and is not — is set out at the end of this appendix.

The elasticity $\times$ operating-cost grid. Table 10 reports the nine-cell grid of Section 4.4 together with the build-vs-slack crossing at each operating cost. The crossing is a fixed-demand object, so it does not vary with

Table 8: KDE discretization of the constructed SAF revenue price

Scenario ω	Probability p^ω	$p_{\text{SAF}}(\omega)$ (\$/gal)
1	0.0328	3.986
2	0.0593	4.224
3	0.0750	4.468
4	0.0749	4.727
5	0.0675	4.999
6	0.0671	5.284
7	0.0743	5.532
8	0.0772	5.821
9	0.0635	6.049
10	0.0444	6.344
11	0.0395	6.610
12	0.0481	6.877
13	0.0581	7.114
14	0.0607	7.400
15	0.0565	7.655
16	0.0454	7.919
17	0.0289	8.168
18	0.0148	8.451
19	0.0079	8.697
20	0.0042	8.944

Twenty equal-width bins of \$0.265/gal spanning the empirical support [\$3.81, \$9.12]/gal, fitted by Gaussian KDE with Silverman bandwidth on the constructed drop-in fuel-plus-RIN price series of Section 4.3 (weekly, August 2013 through December 2021). Bin edges follow from the support and the common width and are not listed, and the cumulative distribution is the running sum of the second column. Probabilities sum to one; $p_{\text{SAF}}(\omega)$ is the bin-conditional expectation, which differs from the bin midpoint by up to \$0.04/gal. The probability-weighted mean is \$6.00/gal and the standard deviation \$1.28/gal.

the elasticity and is constant down each OPEX column; the three distinct values are obtained by re-solving the crossing-region bins at \$1.96, \$2.30, and \$2.70/gal rather than by shifting the baseline crossing one-for-one with operating cost. That shortcut turns out to understate the move: the measured crossings are \$5.91–\$6.06 and \$6.31–\$6.44 against the \$5.8 and \$6.2 a one-for-one shift would give, so the crossing rises 1.1–1.2 times as fast as the operating-cost assumption that drives it (Section 4.4).

Per-scenario realized gaps and asymmetric-gap argument. The 20-bin wait-and-see sweep is solved at a 5% per-scenario relative-gap target; realized gaps cluster around the target but are not uniformly at 5% (Table 12). The median realized gap is 8.07% with inter-quartile range [6.01%, 11.91%], and the maximum realized gap (excluding bin 7, where $\text{PVC}^\omega \approx 0$ mechanically inflates the relative metric) is 21.18%.

The near-zero PVC^ω at bin 7 reflects build-vs-slack indifference at the marginal SAF wholesale price: at $p_{\text{SAF}} = \$5.53/\text{gal}$, the per-gallon welfare contribution of domestic SAF (revenue minus chain cost) is approximately the per-gallon opportunity cost of slack imports ($p^{\text{jet}} = \$3.50/\text{gal}$), so the planner is approximately

Table 9: Computational profile of the Stage-1 LP and Stage-2 MILP at the baseline configuration. The Stage-2 column refers to the cumulative 20-scenario wait-and-see sweep behind the chain-surplus information-value accounting of Section D; per-scenario solves are run at MIP-gap target 0.05; the bin-9 baseline solve realizes an 18.12% gap and the sweep median is 8.07% (the endogenous-demand regret statistics of Section 5.2 come from the separate evaluation of Section 3.5).

Metric	Stage-1 LP	Stage-2 MILP
Continuous variables	489,268	~700,000
Binary variables	0	~10,000
Constraint rows	23,820	~50,000
Non-zeros in constraint matrix	689,597	~3.5 M
MIP gap (5% target)	exact	18.1% realized (baseline)

indifferent between building the marginal facility and purchasing the slack gallon. Bin 7 sits at the structural-equilibrium pivot of the deterministic single-scenario problem, and the relative-gap inflation is a numerical artifact of this indifference, not a methodological failure; the absolute gap \$0.14 B remains within the per-scenario time budget.

Let UB_ω and LB_ω denote, respectively, the solver-reported best-bound and incumbent objective values for wait-and-see scenario ω , and let $w_\omega = UB_\omega - LB_\omega \geq 0$ be the realized absolute gap (the relative gap $g_\omega = w_\omega / |LB_\omega|$ is reported for reference but is sign-unsafe when $LB_\omega < 0$, as at bin 7). By construction $UB_\omega = LB_\omega + w_\omega$ for every ω . Forming $\mathbb{E}[W^*]_{UB} \equiv \sum_\omega p^\omega UB_\omega$ yields an *upper* bound on the true wait-and-see expectation; the corresponding lower bound $\mathbb{E}[W^*]_{LB} = \sum_\omega p^\omega LB_\omega$ would still be valid but does not control the sum of LB_ω from above. What the reported number actually pairs, however, is $\mathbb{E}[W^*]_{LB} = \sum_\omega p^\omega LB_\omega$ — the *incumbent*-weighted wait-and-see expectation, a lower bound on the true $\mathbb{E}[W^*]$ — with $\max_c \mathbb{E}[W | y_c]$, the best-candidate evaluation, itself a lower bound on the true recourse optimum W^{RP*} (the extensive-form recourse problem was not solved). A certified one-sided bound of the form $\mathbb{E}[W^*]_{UB} - W^{RP}_{LB}$ would require the best-bound-weighted expectation $\sum_\omega p^\omega UB_\omega$, which the gap-target budget did not deliver at the low-price bins (median realized gap 8.07%). Because the difference of two lower bounds has no determinate bias sign, we report \$255 M as a *candidate-based estimate* of the chain-surplus EVPI, not as a one-sided upper bound. The same argument shuts down VSS reporting, since W^{EEV} requires solving a separate MILP that carries the same gap asymmetry without a candidate-evaluation upper bound to anchor it (Birge and Louveaux 2011).

The information question on the chain-surplus accounting. The regret of Section 5.2 is, for the fixed central plan, the value of perfect information computed on the endogenous-demand welfare objective. Because the committed plan is fixed rather than recourse-optimal, the \$299 M/yr is an upper bound on the endogenous-demand EVPI of (5) relative to the candidate pool; on the refined evaluation grid the committed plan is itself the pool’s expected-welfare argmax (Section 5.4), so the candidate-based estimate collapses onto the headline figure, while the wait-and-see incumbency gap pushes in the opposite direction, so the figure is not a certified bound. The zero losses in bins 10–20 hold by construction, because the committed plan is the pool’s argmax there, not because those solves closed their gaps; in all twenty bins the recorded wait-and-see value coincides, to machine precision, with the best fixed-siting value in the six-plan pool, so the reported regret is a best-in-pool regret with no wait-and-see content beyond the pool.

Table 10: Nine-cell elasticity $\times$ operating-cost robustness grid

ε	OPEX (\$/gal)	Depots N_D	Refineries N_R	Realized gap (%)	Crossing (\$/gal)
−0.3	1.96	248	34	3.3	5.49–5.54
−0.3	2.30	248	34	3.7	5.91–6.07
−0.3	2.70	248	34	4.1	6.31–6.44
−0.5	1.96	248	34	4.9	5.49–5.54
−0.5	2.30	248	34	5.1	5.91–6.07
−0.5	2.70	248	34	6.4	6.31–6.44
−1.0	1.96	248	34	7.6	5.49–5.54
−1.0	2.30	248	34	11.0	5.91–6.07
−1.0	2.70	185	27	10.6	6.31–6.44

Each cell is a Path B endogenous-demand re-solve at a 5% gap target, warm-started from the nearest cell already solved; the \$2.30 and \$2.70 columns price an H_2 -inclusive configuration (Section 4.4). Bold marks the one cell that departs from the 248/34 first-stage topology; at a 10.6% realized gap it is an unconverged incumbent rather than a certified topology shift. The build-vs-slack crossing is a fixed-demand object and carries no elasticity, so it is constant down each OPEX column. It is reported as an interval, interpolated once from the incumbents and once from the bounds of re-solves of the crossing-region bins at that operating cost: several of those solves stop well short of the gap target, but every bin used to bracket a crossing has its incumbent and its bound on the same side of zero, so its position relative to the crossing is certain and only the interpolated point inherits the gap. Relative gaps are uninformative for bins whose surplus is near zero – the \$2.30 bin at \$6.05/gal reports 817% on an absolute spread of \$0.44 B – and that bin is also where the topology expands from 226/31 to 248/34, so the build threshold and the expansion point nearly coincide.

Table 13 asks the same information question on the fixed-demand chain-surplus (PVC) accounting, a robustness check at a different welfare scale rather than a comparable number. Two caveats govern it. First, the “recourse-problem” entry is not a solved two-stage MILP: it is $\max_c E[W \mid y_c]$ over the twenty wait-and-see candidate plans (the maximizer is the bin-9 plan), a valid lower bound on the true recourse optimum W^{RP*} ; the extensive form was not solved. Second, the wait-and-see expectation is incumbent-weighted ($\sum_\omega p^\omega LB_\omega$), a lower bound on the true $E[W^*]$. The difference of two lower bounds has no determinate bias sign, so the \$255 M/yr figure is a candidate-based estimate of EVPI rather than a certified bound; the asymmetric-gap argument above sets out both bias channels.

The best-candidate (RP-proxy) siting is a strict subset of the union of the 20 wait-and-see (WS) per-scenario optima. All 248 depots placed by the RP-proxy also appear in at least one WS solution; zero are RP-exclusive. The WS union covers 330 distinct candidate locations, of which 82 are WS-only, depots that specific scenarios would prefer to build but that the recourse policy shared across all 20 scenarios elects to forgo. Figure 6 (Section F) panel (a) shows the scenario-recurrence distribution: RP-included depots appear in many WS solutions (mode at $\geq 18/20$), while WS-only depots typically appear in fewer than 5 scenarios. The recourse policy concentrates the chain, picking locations that are robust across the scenario set and deferring scenario-specific marginal capacity to routing flexibility rather than to siting commitment. Panel (b) shows the per-state breakdown: the overlap is preserved across the Southeast core (TN, VA, NC, GA, AL), with the WS-only tail concentrated in the lower-feedstock-density states.

Table 11: Biorefinery capital and operating cost against the public NREL design cases

Study	Pathway	Basis (cost yr)	TCI (\$M)	Output (M gal/yr)	TCI/gal-yr (\$)	Feedst. (\$/dt)
This study	residue FT	FOAK (2022)	947 ^a	50–100	9.5–19 ^a	endog. ^b
Tan et al. (2015)	IDL gasoline	<i>n</i> th (2011)	437.5	47.0	9.31 ^c	80.00
Davis et al. (2018), BDO	biochem.+adipic	<i>n</i> th (2016)	696.8	31.3 ^d	22.26 ^d	71.26 ^e
Davis et al. (2018), acids	biochem.+adipic	<i>n</i> th (2016)	758.4	32.5 ^d	23.34 ^d	71.26 ^e

Notes: Cost years differ across rows and are not deflated to a common basis; the comparison is of order, not of cents. Merrow, Phillips, and Myers (1981) document pioneer-process-plant costs roughly doubling relative to design estimates, supporting the first-of-a-kind (FOAK) multiplier of 1.5–2.5 $\times$ over *n*th-plant totals commonly applied in bioenergy TEA; our \$9.7/gal-yr baseline (97.5 M gal/yr per refinery at the \$947 M anchor) is a $\approx 1.04\times$ premium over the thermochemical *n*th-plant anchor and conservative against that range.

^a Implied per-facility total at the 100 M gal/yr capacity bound (annualized charge \$111.3 M/yr, 20-yr annuity at $\rho = 0.10$); the fixed per-facility charge over the admissible 50–100 M gal/yr band spans \$9.5–\$19/gal-yr, with the baseline solution operating near the upper bound ($\approx$ \$9.7).

^b Stumpage, harvest, and logistics are endogenous model outcomes rather than a delivered-price assumption.

^c Per operating gallon (90% on-stream); the report’s \$8.4/gal-yr figure is per nameplate gallon.

^d Million GGE per year and \$ per annual GGE.

^e Idaho National Laboratory (INL) 2030 target delivered price, not an observed market price. IDL: indirect liquefaction; BDO: 2,3-butanediol intermediate; FOAK: first-of-a-kind.

Table 12: Per-scenario realized MIP gaps on the 20-bin wait-and-see sweep (fixed-demand PVC accounting)

Statistic	Relative gap (%)	Absolute gap (\$B)	Realized PVC ^{ω} (\$B)
Min (excl. bin 7)	4.75	0.02	–
Median (excl. bin 7)	8.07	0.13	–
Max (excl. bin 7)	21.18	0.46	–
Bin 7 (separately)	455.80 [†]	0.14	–0.03

Notes: Summary statistics exclude bin 7 ($p_{\text{SAF}} = \$5.53/\text{gal}$), whose near-zero $\text{PVC}^{\omega} = -\$0.03$ B mechanically inflates the relative gap (455.8%); its \$0.14 B absolute gap remains within the per-scenario time budget and is not anomalous in absolute terms ([†]).

Cross-checking the closed form against the MILP. The MILP disciplines the closed form’s two free cost parameters. At interior optima the wait-and-see solves trace the toy’s first-order condition $w_{\omega} = c + K Q_{\omega}$, so regressing the per-bin demand intercept on realized wait-and-see output identifies both jointly: across all 20 bins, $K = 4.3 \times 10^{-9}$ \$/gal² and $c = \$5.00/\text{gal}$ ($R^2 = 0.68$), with implied capacity curvature $k = K - b = 3.1 \times 10^{-10}$, small relative to the demand slope. Subsample fits are unstable — bins 1–9 give $K = 2.0 \times 10^{-9}$, and bins 10–20, where output is recourse-constrained by the fixed 248/34 topology, admit no sensible fit — which is direct evidence of the integer kinks that make the MILP’s effective curvature piecewise rather than constant; we carry that instability as the calibration’s uncertainty band.

Three checks follow. First, the *contraction*: in the collapsed regime the $\lambda = 1$ optimum has effective intercept A_L (Section A, equation (34)), so the tail-only solution is the deterministic optimum at the low state. With the calibrated c , the predicted contraction in the endogenous-demand intercept space is $(w_{\text{bin1}} - c)/(\bar{w} - c) = 0.84$, against the realized bin-1 wait-and-see contraction of $192/248 = 0.774$ on depots and $26/34 = 0.765$ on refineries: the MILP contracts more than the smooth prediction, consistent with minimum-throughput lumpiness shedding whole facilities at the margin. Second, an *information-value reconciliation*: $\text{EVPI} = \text{Var}(A)/(2K) = \190 M/yr

Table 13: Information-value accounting on the chain-surplus (PVC) objective at the deterministic 248-depot/34-refinery baseline

Quantity	Value (\$ million)
$\sum_{\omega} p^{\omega} \text{LB}_{\omega}$, incumbent-weighted WS expectation (lower bound on $\mathbb{E}[W^*]$)	1,541
$\max_c \mathbb{E}[W \mid y_c]$, best candidate plan (lower bound on $W^{\text{RP}*}$; extensive form not solved)	1,286
W^{EEV} value of expected-value solution	n.r.
EVPI , candidate-based estimate (difference of the two rows above)	255 (19.8%)
VSS ($= W^{\text{RP}} - W^{\text{EEV}}$)	n.r.

Notes: All quantities are single-year-equivalent surpluses (\$M/yr; capital as a 20-year annuity at $\rho = 0.10$), computed on the fixed-demand model (Section 3.5); the EVPI percentage is relative to the best-candidate RP-proxy row (255/1,286). They are not commensurable with the endogenous-demand regret statistics of Section 5.2. Both reported rows are lower bounds on their true objects (the WS expectation is incumbent-weighted; the RP row is a best-candidate evaluation in the spirit of the candidate-evaluation literature, Mak, Morton, and Wood 1999; Bayraksan and Morton 2011, though we do not run their replication schemes), so their difference is a candidate-based EVPI estimate with undetermined bias sign, not a certified bound. EEV-based rows and VSS are not reported (n.r.) because they would require a further nonzero-gap MILP without an anchoring bound.

at the calibrated curvature, the same order as the \$255 M/yr candidate-based chain-surplus estimate and the \$299 M/yr best-in-pool regret. The three accountings differ in construction and are not expected to coincide, but they agree in magnitude, which is what Corollary 2 can deliver on a piecewise-curvature problem. Third, the *benefit–cost content* of Corollary 4 reported in Section 5.6: the recovered deadweight loss $\sigma^{*2}/(2K) = \$31\text{--}\65 M/yr sits an order of magnitude below the flat credit’s MCPF cost. One arithmetic identity links the regret and expectation findings of Section 5.4 and should not be double-counted: because the certified wait-and-see column coincides with the pool maximum in every bin, the \$99.5 M/yr expected-welfare margin of the committed plan over the 197/27 plan equals the expected-regret difference \$400 – \$300 M/yr; the two rankings share one expected-value number and part ways only in the tails.

Two-state injection test: free solves and caveats. The injection test of Section 5.4 runs in two layers. In the first, free extensive-form solves at $\lambda = 0$, at $\lambda = 0.5$, and at $\lambda = 0.5$ with a \$0.518/gal credit all return the committed 248/34 topology at realized gaps of 3.0–4.1%: the risk perturbation’s predicted output effect ($\sigma^*/K \approx 0.12$ B gal, 3.7% of output; a welfare effect on the order of 0.1–0.2%) sits below the resolvable tolerance of the free MILP, which is the precise sense in which the joint mean–CVaR program remains open. The second layer fixes the sitings, which makes the test exact: LP evaluations of the committed plan and the two leading contracted candidates under the same three configurations, at the refined 81-breakpoint grid, resolve the ranking to machine precision and give the 94% restoration share reported in the text (97% at the coarse grid). Two caveats keep the test in its lane. Collapsing twenty bins to two compresses the high-price upside that favors the large plan, so plan *levels* are not transportable from the collapse to the full bin set, where the committed plan leads expected welfare (Table 6). And the fixed-demand model’s interpolated build-vs-slack crossing sits at $\approx \$5.5/\text{gal}$, so the low-price states are near the no-build margin, whereas Assumption 5 keeps low-state production interior; the closed form’s per-gallon policy objects (σ^* , the floor strike, the outlay ratios) are first-order summaries away from that margin, not at it. The closed form predicts $\sigma^*(\lambda)$ linear in λ in the collapsed regime with slope

$\pi_H(w_H - w_L) = 0.472 \times 2.194 = \$1.04/\text{gal}$ per unit λ ; whether the full 700k-variable MILP at intermediate λ would deliver the same linearity, or exhibit integer kinks at depot-count transitions that the two-state model abstracts away, is the open question recorded in Section 6.3.

E. Section 45Z: Institutional Detail

This appendix records the institutional machinery behind the §45Z clean fuel production credit that our decomposition takes as its policy benchmark, and the respects in which certified credit values could differ from the GREET/CORSIA-derived carbon intensities used in the model.

Carbon-intensity certification. The credit equals an applicable amount times an emissions factor, $(50 - \text{ER})/50$, where ER is the fuel’s lifecycle emissions rate in kgCO₂e per mmBtu ([United States Code 2025a](#)). For fuel produced from 2025 onward, Treasury guidance (Notices 2025-10 and 2025-11, issued January 10, 2025) directs producers to the 45ZCF-GREET model, a purpose-built derivative of Argonne’s GREET model published by the U.S. Department of Energy on January 15, 2025 and updated in May 2025 ([Internal Revenue Service 2025a,b](#); [U.S. Department of Energy 2025](#)). The model covers gasification–Fischer–Tropsch pathways, but industry comment letters and trade analyses indicate that forestry and logging residues are not among its listed feedstocks, so a forest-residue FT producer must instead petition for a provisional emissions rate (PER): the producer first obtains an “emissions value” from DOE (requiring a front-end engineering and design study; DOE published its notice of intent for this process on July 24, 2025), then files a PER petition with the Internal Revenue Service (IRS); Treasury has indicated credits may be claimed retroactively to January 1, 2025 once a rate is established ([RSM US LLP 2025a](#); [Strategic Biofuels LLC 2025](#); [Advanced Biofuels USA 2025](#)). Our carbon-intensity range of 8.6–12.9 kgCO₂e/mmBtu is GREET/CORSIA-derived; the certified 45ZCF-GREET or PER value could differ, since taxpayers may not substitute their own background assumptions ([Holland & Knight LLP 2025](#)). At our range, the emissions factor is 0.74–0.83, so the credit per gallon is that fraction of the applicable amount below. Proposed regulations issued February 4, 2026 would further specify these rules ([U.S. Department of the Treasury and Internal Revenue Service 2026](#)).

Rates and the 2025 amendments. As enacted in the Inflation Reduction Act, §45Z paid a base rate of \$0.20/gal (\$1.00/gal if prevailing-wage and apprenticeship requirements are met), with a SAF premium rate of \$0.35/gal (\$1.75/gal) ([United States Code 2025a](#); [Congressional Research Service 2025a](#)). The 2025 budget reconciliation act (P.L. 119-21, enacted July 4, 2025) eliminated the SAF premium for fuel produced after December 31, 2025 — aligning SAF at the \$1.00/gal maximum — extended the credit through December 31, 2029 (from 2027), restricted eligible feedstocks to those produced or grown in the United States, Mexico, or Canada, excluded indirect land-use-change emissions from the emissions rate, and disallowed negative emissions rates except for animal-manure fuels ([RSM US LLP 2025b](#); [Clean Air Task Force 2025](#)). The dollar amounts are indexed for inflation for years after 2024 ([United States Code 2025a](#)). The §40B SAF credit that preceded 45Z (\$1.25–\$1.75/gal in 2023–24) expired December 31, 2024 ([Internal Revenue Service 2024](#)).

Transferability. Because SAF developers rarely have tax appetite in early operating years, most 45Z value is monetized by selling credits for cash to unrelated taxpayers under §6418 ([United States Code 2025b](#)). Market

intelligence for 2025 reports transferable credits trading at 92–94 cents per dollar on average, with §45Z credits pricing *below* the market average — roughly 91 cents — reflecting their recent emergence, verification burden, and guidance risk (Crux Climate 2025). This discount is one concrete channel through which a statutory dollar of flat credit delivers less than a dollar of financeable revenue, whereas floor payments from a creditworthy counterparty enter lenders’ cash-flow models at face value.

Stacking with the RFS and state LCFS programs. FT jet fuel from forest residues qualifies under the Renewable Fuel Standard as cellulosic diesel (a category that includes jet fuel), generating D-code 7 RINs; equivalence values for renewable jet fuel are assigned facility-by-facility under 40 C.F.R. §80.1415(c) and have historically been approved at 1.6–1.7 RINs per gallon, with most at 1.6 (Code of Federal Regulations 2025a,b). California, Oregon, and Washington treat alternative jet fuel as an *opt-in* fuel in their low-carbon fuel standards — SAF generates credits while conventional jet fuel generates no deficits — and these state credits stack with RINs and 45Z (California Air Resources Board 2024; Stillwater Associates 2025). Within the tax code, stacking is restricted: no 45Z credit is allowed for a facility for which the §45V clean hydrogen or §45Q carbon sequestration credit is allowed in the same taxable year, determined year by year (United States Code 2025a; Baker Botts LLP 2025). Capture differs across instruments: 45Z pays the registered *producer* (registration under §4101; sale to an unrelated party) (Holland & Knight LLP 2025), whereas the expired §40B credit attached to the qualified SAF *mixture* and was therefore captured at the blender level (Internal Revenue Service 2024); RINs are generated at production but separated and monetized upon blending, splitting value between producer and obligated party.

Incidence and risk-bearing. Statutory incidence of 45Z at the producer matters for our decomposition because the instrument’s risk-mitigation value depends on who holds the price exposure it is meant to offset. A flat per-gallon credit raises the mean of the producer’s revenue distribution but leaves its variance untouched; if offtake agreements pass price risk through to the producer (as in index-linked SAF offtakes), the risk component of the credit identified in Section 5 compensates the party that actually bears the tail risk, which is the efficiency argument for converting that component into a floor. If instead airlines absorb price risk through fixed-price offtakes, a producer-side flat credit over-delivers to the party bearing the least risk. The producer-level payment point of 45Z — unlike the blender-level 40B — therefore makes the floor conversion administratively natural: the same registered entity that certifies volumes and carbon intensity for the credit would be the counterparty of the floor contract.

Contract template for a SAF price floor. The floor priced in Section 5.6 is implementable with existing contract technology, but not through the tax code alone. *Settlement index:* no deep SAF spot market exists. Argus and Platts publish U.S. SAF assessments (Gulf Coast, Illinois, West Coast), but the market remains thin enough that these are indicative rather than transaction-based (Argus Media 2025; S&P Global Commodity Insights 2025; Airlines for America 2025), so a workable contract would settle on a liquid Gulf-Coast jet-A assessment plus a SAF adder that is either fixed administratively or discovered at auction. *Template:* the U.K. Contracts-for-Difference scheme shows both the mechanics and the strike-setting risk. Strikes are discovered in pay-as-clear auctions capped by administrative strike prices, with a 15-year tenor, CPI indexation, and a government-owned counterparty (the Low Carbon Contracts Company, LCCC) (Low Carbon Contracts Company 2025). Allocation Round 4 (2022) cleared offshore wind at £37.35/MWh (2012 prices); Round 5 (2023) set the administrative cap at

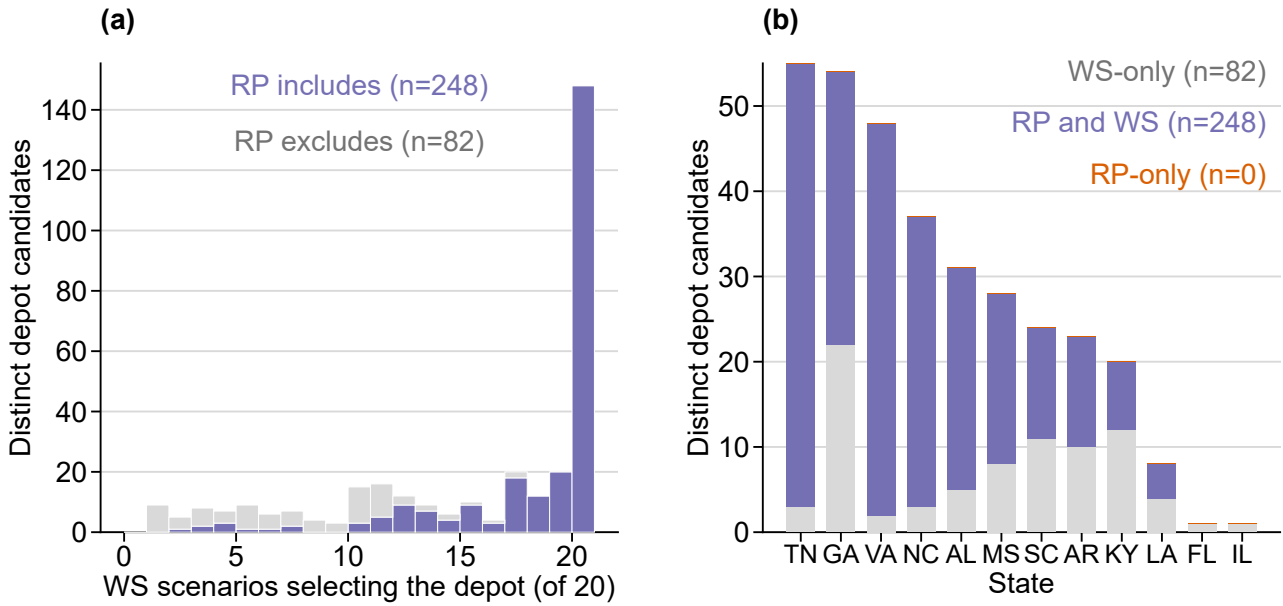


Figure 6: Siting comparison: recourse-problem first-stage solution (RP, 248 depots) vs. union of 20 wait-and-see per-scenario solutions (WS, 330 depots union). **Panel (a):** scenario-recurrence histogram — for each depot in the WS union, the number of WS scenarios (out of 20) in which the depot appears, separated by whether the RP solution includes the depot (purple) or excludes it (light gray). **Panel (b):** per-state stacked breakdown of WS-only / both / RP-only depot candidates across the SE-U.S. core and broader region.

£44/MWh and attracted zero offshore-wind bids after costs rose; Round 6 (2024) raised the cap to £73/MWh and cleared at £58.87/MWh (Regen 2022; Offshore Renewable Energy Catapult 2023; UK Department for Energy Security and Net Zero 2024). Auctions reveal costs, but administered caps set on stale information can misfire in either direction, which bears directly on setting a SAF strike from TEA estimates. *Required elements:* a tenor matched to project debt (10–15 years), a per-facility volume cap, CPI indexation of the strike, competitive strike discovery with an administrative reserve, an explicit allocation of SAF-versus-jet-A basis risk (settling on jet-A leaves the SAF premium with the producer; settling on a SAF assessment imports thin-index risk), and a creditworthy counterparty. Because Treasury’s credit machinery is ill-suited to contingent two-sided payouts, natural administrators are the DOE Loan Programs Office’s project-finance apparatus or a dedicated federally backed counterparty analogous to the LCCC (U.S. Department of Energy, Loan Programs Office 2025). Auction administration, counterparty capitalization, and monitoring absorb resources, and the \$0.48 B/yr saving reported in Section 5.6 is gross of these administrative and transaction costs.

F. Additional Figures

This appendix collects three supporting figures referenced in Section 5: the siting comparison behind the information-value accounting of Section D (Figure 6), the implied-SCC schedule behind the emissions-basis discussion of Section 5.5 (Figure 7), and the fixed-demand topological price sweep of Section 5.1 (Figure 8).

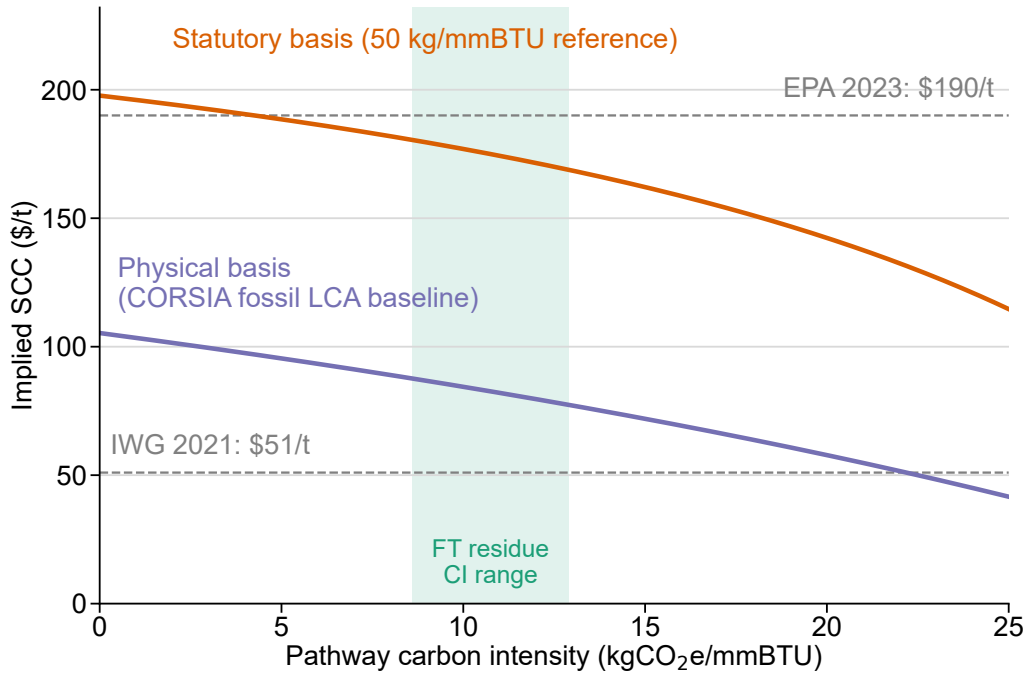


Figure 7: Implied SCC as a function of pathway carbon intensity. Both the statutory credit, $1.75 \times (50 - \text{CI})/50$, and the emissions differential move with CI, so the implied SCC $(\text{credit}(\text{CI}) - \sigma^*)/\Delta e(\text{CI})$ is a schedule. The lower (purple) schedule is the physical basis ($\Delta e = 11.70 - 0.1246 \text{ CI kgCO}_2\text{e/gal}$, CORSIA fossil life-cycle baseline); the upper (orange) schedule is the statutory basis ($\Delta e = 0.1246 (50 - \text{CI})$, the statute's $50 \text{ kgCO}_2\text{e/mmBTU}$ reference). Shaded band: the FT-residue CI range $8.6\text{--}12.9 \text{ kgCO}_2\text{e/mmBTU}$ consistent with the $\$1.30\text{--}\$1.45/\text{gal}$ credit. The point band $\$77\text{--}\91 in (27) holds Δe fixed at 10.2 while varying the credit; the schedule here co-moves both, giving $\$77\text{--}\88 on the physical basis over the shaded range. Horizontal lines: the two federal SCC vintages.

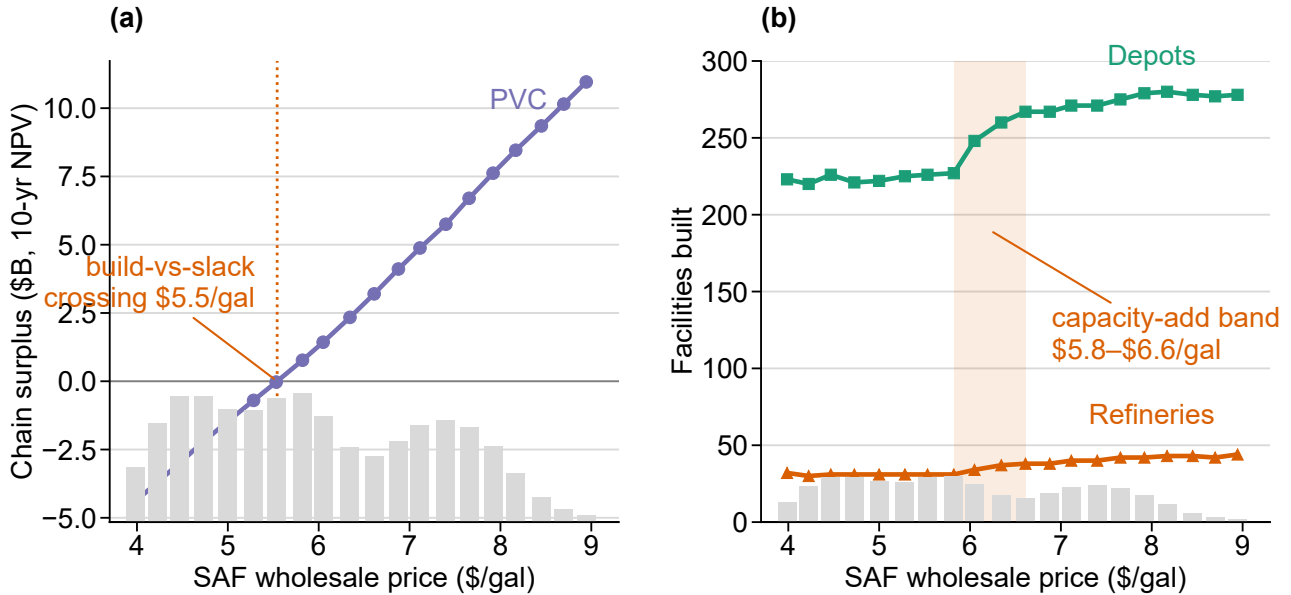


Figure 8: Topological response of the SE-U.S. residue-to-SAF supply chain to wholesale SAF price, across the 20 KDE-discretized scenarios (gray bars: scenario probabilities p^ω from Table 8). **Panel (a):** present-value chain surplus $\text{PVC}(\omega)$ vs. $p_{\text{SAF}}(\omega)$; the dotted orange line marks the interpolated build-vs.-slack crossing at $\approx \$5.5/\text{gal}$, which falls between the $\$5.53$ bin ($\text{PVC} = -\$0.03\text{B}$) and the $\$5.82$ bin ($+\$0.77\text{B}$). **Panel (b):** realized facility counts N_D, N_R vs. p_{SAF} ; the shaded band is the marginal-capacity expansion range, roughly $\$5.8\text{--}\$6.6/\text{gal}$, across which the chain adds depots and refineries quickly.